EIGSEP Memo 005

Marjum Pass 2026-07: beam reconstruction from the transmitter raster

EIGSEP Team

Living document. This version: 2026-11-09

Abstract

How box-air's beam is measured from the 07-17 transmitter raster, and how well. The transmitter's comb teeth, seen through box-air as it rolls and tips, sample the beam along the raster's track (41% of HEALPix pixels), each tooth in one of the transmitter's two orthogonal polarizations. This memo describes the data selection, the background under each tooth, the beam model (spherical harmonics times a smooth spectral basis, regularized toward the HFSS simulation), the geometry fit and the held-out test, and states how far the reconstructed beam improves on HFSS. The reference product is `derived/beam/empirical_raster_v0018`, fitted on pointing table v6, whose azimuth is left-handed about the boresight (memo 003, §5.1), with the antenna and transmitter at memo 002's adopted `v0005` posterior means and the tilt of the elevation axle along the highline held at the box IMU's mean over the raster, $+5.91°$ (memo 003, §7). Its DPSS penalty is held at 10^{-2}: at the weaker penalty the validation scores prefer (v0017's fit, as `empirical_raster_v0016`), the exported beam near 66.4 MHz has a lobe 59° from the boresight at 5.2× HFSS's peak, which the held-out scores do not see. v0018 replaces v0017, which fitted the tilt ($+4.74°$) on pointing table v5. Fitted again on v6, the tilt stays near $+4.74°$ but predicts held-out data no better than the IMU's (per tooth, the difference lies within twice its jackknife error on 56 of 64 teeth): the raster cannot separate a change of tilt from a small change of the beam's shape, and the fitted value says nothing about the mount. The antenna is bolted to the mount on the box, so the axle tilt is the IMU's (§5.6). This memo does not refit the beam: every number, figure and table is computed by `make_products.py` in this directory from the products' per-sample data and predictions (Appendix A). §5 compares v0018 with v0017, the previous release, and with the fitted-tilt variant. §7 checks the HFSS beam's frequency interpolation against a 1 MHz solve of the same model. Pointing, the mount and the azimuth zero are in memo 003.

1 Summary

- **Data**: 64 transmitter teeth from 64.5 to 240.2 MHz (fit band 64.2–240.5 MHz; 25 on arm 0, 39 on arm 1), 6,579 samples over 20:26:00–21:28:40 UTC on 07-17 at the ∼91m height era, split by 12° azimuth stripes into training (4,324), validation (1,047) and test (1,208) samples.
- **The empirical beam beats HFSS on the test stripes** (§6): fractional rms 0.077 against 0.094 (means over teeth weighted by sample count), better on 45 of 64 teeth, with the worst tooth at 0.246 against 0.276 (Table 9). v0017 scored 0.076 against 0.091 on its own test samples.
- **The residual is structure, not noise**: it is ∼6× the white-noise floor (0.014). What is left is in the model, the background or the geometry.
- **Geometry**: azimuth is left-handed (memo 003, §5.1). The fit puts the table's zero $+15.97°$ from the highline, as memo 003 finds from the arm ratio alone, and transmitter arm 0 at bearing 172°. The beam cannot be tipped along the highline by the az and el zeros; a tilt of the elevation axle does it, with the transmitter held at memo 002's position. The box's IMU measures the tilt per sample (memo 003, §7); v0018 holds it at the mean over the fit's 6,328 kept samples, $+5.91 \pm 0.15°$. Fitted instead, it comes out $+4.74 \pm 0.23°$, as in v0017, but predicts the held-out stripes no better: v0018's held-out fractional rms minus the variant's is $-2.3 \pm 5.3 \times 10^{-4}$ on average over teeth (jackknife over the 4 validation stripes), within twice its error on 56 of 64 teeth (Fig. 3). Only the HFSS beam prefers the fitted tilt: the raster cannot separate a change of tilt from a small change of the beam's shape, and the fitted value says nothing about the mount, to which the antenna is bolted on the box (§5.6). The fitted-tilt variant is kept as the

documented alternative: v0018's beam moves from v0017's by 0.074 dB rms (0.28 dB within 20° of the boresight), the variant's by 0.014 dB.

- **v0018 against v0017**: pointing table v6 adds 57 samples within 7.0–9.9° of the boresight, and raster flags v0002 cover them. On the validation samples the two share, which by construction exclude those rows, the held-out fractional rms changes by $-2.9 \pm 5.1 \times 10^{-4}$ on average (better on 38 of 64 teeth), and the exported beam by 0.074 dB rms on the sampled directions and at most 0.28 dB within 20° of the boresight, mostly from the tilt (Figs. 3 and 4).
- **Penalty**: the validation scores cannot choose the DPSS penalty. At 10^{-3}, where 10^{-2} is 4.2% worse on validation, v0017's exported beam (as `empirical_raster_v0016`) at 66.4 MHz differs from HFSS by 3.8 dB rms on the sampled pixels and its peak, 59° from the boresight, is 5.2× HFSS's. In v0018, at 10^{-2}, these are at most 0.4 dB and 1.0× below 90 MHz (Table 2). The penalty is held at 10^{-2}.
- **HFSS at 1 MHz** (§7): the HFSS model solved at every integer MHz from 50 to 250 MHz confirms that the 52-frequency beam is smooth enough in frequency. Interpolated linearly, as in the beam fitter, it reproduces the 1 MHz beam to at most 0.28% of the peak power in any direction (the held-out residual is at least 3.9%); with four-node weights, to 1.1×10^{-4}. The 201-frequency beam is released as `derived/beam/hfss_bowtie_1mhz_v0001`.
- **Limits**: power-only data leave the beam's phase undetermined. Each tooth is one polarization, and below 132.8 and above 230.5 MHz only arm-1 teeth are fitted, so there the raster measures one projection of the field (§2). Directions the raster never sampled (41% of HEALPix pixels are sampled, counted from v0018's export) are set by the HFSS prior and the penalty, and no test reaches them (§8).

2 Data

Samples. Box-air, antenna-input rows of the elevation-sweep raster, 20:26:00–21:28:40 UTC on 07-17 (memo 003, §4), with pointing-table v6 quality `ok`: commanded azimuth, left-handed about the boresight, IMU elevation, and the axle's tilt (memo 003, §7). Only rows in the ∼91m height era are kept (236 dropped; `curation/mode_table.jsonl`), and isolated elevation-solution glitches are dropped (2). Throughout the raster box-air rolls about its boresight in 5° steps while the elevation axis turns through full revolutions, so the transmitter is seen at every angle from the boresight and at a range of roll angles.

Background under each tooth. The comb sits on box-air's sky and receiver spectrum. For each integration a robust DPSS model is fitted to the spectrum in 48 MHz windows with a 150 ns delay half-width, masking each tooth's channel and the channels either side, and the FM band (87–108.5 MHz), with 4σ clipping; the tooth signal is the measured power minus this background. The background keeps every DPSS mode with concentration $\geq 10^{-6}$ (21 modes); a floor$(2NW) + 1$ rule could not represent a constant near the window edges (`eigsep_data.beam_mapping.tx_background`). Broadband dropouts are masked by a gap-ratio test, and `raster_flags_v0002` removes 251 samples and 34 tooth spikes (§5.2); it covers every sample.

Which teeth. The candidates are the 101 channels $\equiv 0 \bmod 8$ inside the HFSS frequency range. A tooth is kept if it is isolated (the 90th percentile of its signal-to-background ratio over the raster is at least 10 times the larger of its background error in a transmitter-off control interval, 17:03–17:54, and its local residual), if at least half its samples are good, and if it is coherent with its neighbours: its log signal must correlate at ≥ 0.9 with the mean of the two nearest selected teeth on the same transmitter arm, iterated by dropping the least coherent tooth (`tooth_selection_v0002`). 14 fail isolation, 11 are in the FM band, 10 fail coherence and 2 have too few good samples (Fig. 1). The coherence test removes teeth with intermittent RFI that pass isolation. Of the teeth kept, 39 are on arm 1, from 64.5 MHz, and 25 on arm 0, from 132.8 to 230.5 MHz; the two arms are the transmitter's orthogonal linear polarizations (Eq. 1). Below 132.8 MHz, and above 230.5 MHz (the top 3 teeth, 232.4–240.2 MHz), only arm-1 teeth are fitted, so there the raster measures one polarization projection of the field in each direction; the orthogonal component, and with it part of the power beam, comes from the fit's prior toward HFSS and the spectral basis's link to the teeth where both arms are fitted. The prior is the loss term $\lambda \|\mathbf{C} - \mathbf{C}_0\|^2 / \|\mathbf{C}_0\|^2$ toward the HFSS coefficients, with $\lambda = 0.01$ in v0018 (§3; `eigsep_data.beam_mapping.tx_fit`, `JointBeamFit.fit`).

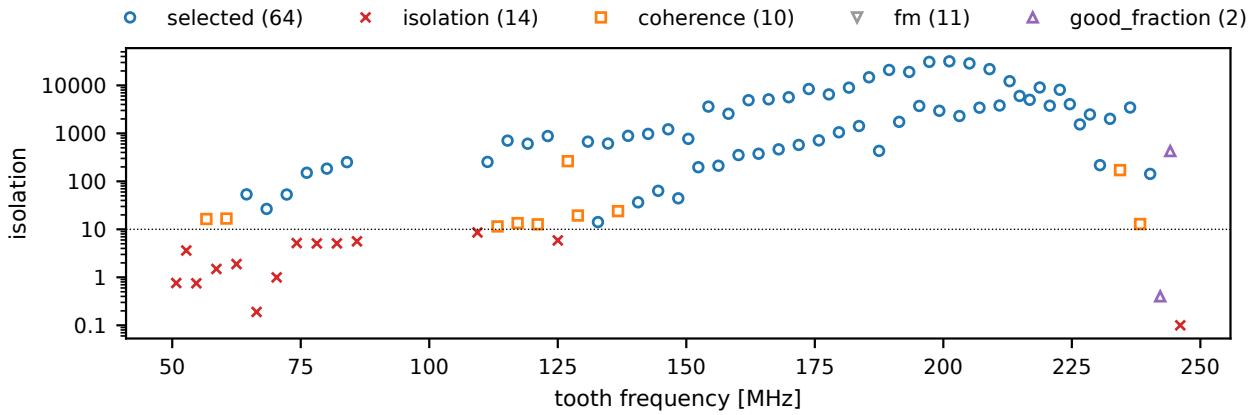


Figure 1: The selection keeps 64 of the 101 candidate teeth: 14 fail isolation, 11 are in the FM band, 10 fail coherence and 2 have too few good samples. Tooth isolation against frequency, marked by the reason each candidate was kept or dropped. The dotted line is the isolation threshold.

3 Model

Coupling. Box-air is fixed in position relative to the transmitter, so the transmitter's own gain toward box-air is the same for every sample and is absorbed, with the drive and the cable losses, into one positive gain g_f per tooth. The model power for tooth f, driven on transmitter arm $a(f)$, is

$$P_f(t) = g_f \left|\mathbf{E}_f(\theta_t, \phi_t) \cdot \hat{\mathbf{e}}_{a(f)}(t)\right|^2, \tag{1}$$

where (θ_t, ϕ_t) is the transmitter's direction in box-air's body frame (memo 003, Eq. 1), $\hat{\mathbf{e}}_a$ is the field direction of transmitter arm a (horizontal dipoles at polarization angle α and $\alpha + 90°$) expressed in the same frame, and $\mathbf{E}_f$ is box-air's complex vector far-field pattern. Both arms see the same beam through orthogonal polarizations.

The simulated beam. Dominic's HFSS model of box-air's bowtie antenna gives $\mathbf{E}_f$ as Cartesian field components on a HEALPix grid (nside 32) at 52 frequencies, 50.8–250.0 MHz (`eigsep_data/hfss_beam_maps/bowtie_beam.npz`). The fields are normalized to unit mean power at 190 MHz with the common phase removed, and interpolated linearly to the teeth. A 1 MHz solve of the same model shows that interpolation costs at most 0.28% of the peak power (§7).

The empirical beam. Each Cartesian component is a sum of real spherical harmonics to $\ell_{\max} = 8$ (81 terms) with coefficients that are themselves smooth in frequency, expanded in a spectral basis: $\mathbf{E}_f(\theta, \phi) = \sum_k s_k(f) \sum_{\ell m} \mathbf{C}_{k\ell m} Y_{\ell m}(\theta, \phi)$. Two spectral bases are fitted: 8 DPSS modes with a 12.5 ns delay half-width (sinc-continued across the band), and 5 principal components of the HFSS fields. $\ell_{\max} = 8$ represents the HFSS fields to 0.0039 worst-case relative error, and the two spectral bases to under 1%. The loss is the mean over teeth of the squared residual, each tooth scaled to unit rms, plus a prior $\lambda \|\mathbf{C} - \mathbf{C}_0\|^2 / \|\mathbf{C}_0\|^2$ toward the HFSS beam's own coefficients $\mathbf{C}_0$; the gains are fitted in log so they stay positive (JAX, L-BFGS-B; `tx_fit.JointBeamFit`).

Training, validation and test. Samples are assigned by stripes of commanded azimuth 12° wide: stripe k is validation if $k \bmod 5 = 1$, test if $k \bmod 5 = 3$, and training otherwise. The beam is fitted on training samples at $\lambda = 10^{-2}, 10^{-3}, 10^{-4}$. Validation scores differ between penalties by a few per cent (Table 1), and the weakest may stop at the iteration limit, so each basis takes the strongest converged penalty whose validation score is within 1% of the best. That rule chooses 0.01 for PCA. For DPSS it chooses 10^{-3}, against which 10^{-2} is 4.2% worse on validation, but the penalty is held at 10^{-2} (Table 1; `fit_beam.py --penalty dpss=1e-2`). In `empirical_raster_v0016`, v0017's fit at 10^{-3}, the exported beam at 66.4 MHz has a lobe 59° from the boresight at 5.2× HFSS's peak, and its total power differs from HFSS's by 3.8 dB rms even on the sampled pixels (Table 2), yet the held-out scores do not see it: of v0018's fits on the training stripes (Table 1), the weaker penalty's test score below 87 MHz is the better. The data measure the power in the transmitter's polarization

along the raster's track, not the total power or the directions off it, so the scores cannot rule such a beam out; the stronger penalty keeps it near HFSS there. The beam fitted on the training stripes at the chosen penalty is the *held-out* beam, whose test score is an independent measure. The chosen fit is then refitted on all samples and exported in the HFSS file format (*refit* and *exported*; their scores are in-sample, and the export changes the predictions by at most 0.20%). Outside 64.2–240.5 MHz the exported slices are the normalized HFSS beam.

Table 1: DPSS penalty selection: each penalty's validation median per-tooth fractional rms and whether it converged, and the count-weighted test fractional rms by band of the beam fitted on the training stripes at that penalty.

DPSS penalty λ	10^{-2}	10^{-3}	10^{-4}
validation median rms	0.0672	0.0645	0.0650
converged	yes	yes	no
test fractional rms, band [MHz]			
50–87	0.066	0.061	0.063
87–145	0.082	0.083	0.083
145–190	0.073	0.073	0.074
190–235	0.082	0.079	0.078
235–245	0.073	0.066	0.067

Table 2: The exported DPSS beam against HFSS at each HFSS frequency below 90 MHz, and the worst above it, for v0018 (DPSS penalty 10^{-2}) and for `empirical_raster_v0016`, v0017's fit at the 10^{-3} the selection rule chose (v0018's beam differs from v0017's by 0.074 dB rms on the sampled pixels, §5). Each total-power map is scaled to unit visit-weighted mean on the sampled pixels. "rms" is the visit-weighted rms of the dB ratio on the sampled pixels; "peak ratio" is the ratio of the two maps' maxima over the whole sphere.

	`empirical_raster_v0016`, $\lambda = 10^{-3}$		`empirical_raster_v0018`, $\lambda = 10^{-2}$	
MHz	rms [dB]	peak ratio	rms [dB]	peak ratio
66.4	3.8	5.25	0.4	0.98
70.3	2.5	2.29	0.4	0.98
74.2	1.7	1.30	0.4	1.00
78.1	1.2	1.07	0.4	0.99
82.0	0.8	1.01	0.3	0.98
85.9	0.5	0.97	0.3	0.98
89.8	0.5	0.99	0.3	0.97
≥90, worst	0.9	1.13	0.9	1.21

4 Geometry

The positions are memo 002's adopted `v0005` posterior means (antenna in the 91 m era, transmitter; `memos/memo-002-marjum-2026-07-geometry/tables/summary.json`, which this memo checks against the product's provenance); the mount and convention are memo 003's, with $\psi = 142.164°$ and the table's azimuth left-handed about the boresight (pointing tables v5 and v6). Two quantities are not known in advance: the azimuth offset added to the table's azimuth, and the transmitter's arm-0 angle α (the arm-0 field is $(-\sin\alpha, \cos\alpha, 0)$ in ENU, a compass bearing of $180° - \alpha$). A grid search with the HFSS beam finds them together (Fig. 2): offset $+16°$ and $\alpha = 8°$, normalized rms 0.119, against 0.328 at the best α with zero offset. v0017's scan found the same offset and α (Table 5).

Axle tilt. Small corrections are then fitted on top in nested sets, and the set with the best validation score is kept (Table 3). The azimuth and elevation zeros turn the beam about the boresight and about the elevation axle, which runs along the highline; neither can tip the beam along the highline. Transmitter offsets can, but fitted they ran to their bound along the highline (in v0012 and v0013, on pointing table v2.0; Table 10). The transmitter stays where memo 002 puts it, and the sixth correction is a tilt of the elevation axle out of the horizontal, a

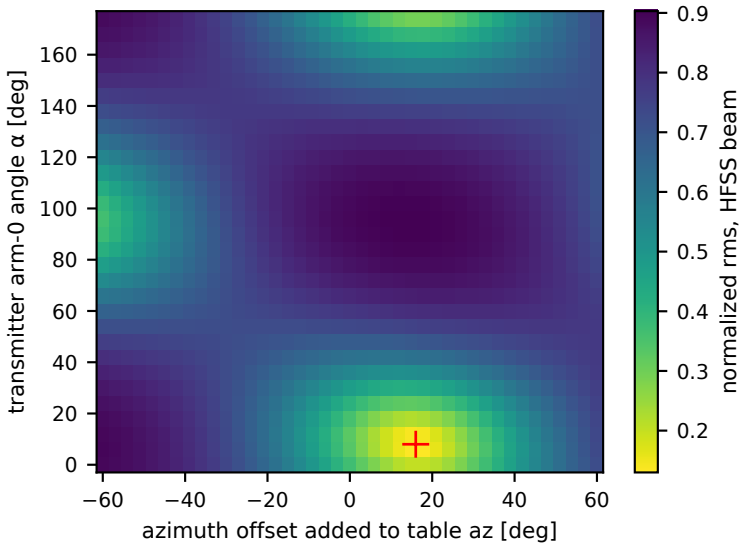


Figure 2: The HFSS beam fits best at an azimuth offset of +16° and $\alpha = 8°$ (normalized rms 0.119). Its fit to v0018's training samples over the azimuth offset and the transmitter arm-0 angle α, with no corrections (so no axle tilt), as normalized rms; the cross marks the minimum.

rotation about the horizontal axis perpendicular to the highline (the mount becomes $R_z(\psi)\,R_y(t)\,R_x(\mathrm{el})\,R_z(\mathrm{az})$; positive t lowers the ψ end; `eigsep_data.beam_mapping.tx_fit`). Fitted on v0018's samples with t free (the fitted-tilt variant), the best validation score without the tilt is 0.0903; with it, 0.0758 (`axle_tilt`, condition number 1.0).

The IMU's tilt. Box-air's elevation IMU rides on the box, and the gravity component along the elevation axle gives the axle's tilt per sample (memo 003, §7); pointing table v6 carries it as `tilt_deg`. v0018 holds t at its mean over the fit's 6,328 kept samples, $\tau_{\mathrm{IMU}} = +5.91 \pm 0.15°$: the error is memo 003's systematic for the axle's direction, and the samples scatter about the mean by 0.89°. One t serves the whole raster; the mount does not follow the tilt sample by sample. With t held, the kept corrections are `polarization_az_el`: $\delta\mathrm{az} = -0.03°$, so the table's zero lies +15.97° from the highline, $\delta\mathrm{el} = +0.09°$ and $\delta\alpha = +0.08°$, with a validation score of 0.0792 (Tables 3 and 5). Why v0017's fitted +4.74° differs from the IMU's, and why v0018 does not fit the tilt, is §5.6.

Handedness. Memo 003 shows that the azimuth is left-handed about the boresight, from `imu_az`'s turn with the shaft, and that the arm ratio alone cannot tell the handedness. The beam pattern breaks the mirror symmetry only through the tilt and the beam's own asymmetries, so this fit is not an independent test. Rerunning the geometry stage on v0018's data with the table's azimuth negated (right-handed; Table 4) gives the mirrored zero (−16°) and arm 0 at 84°. With the tilt free the two handednesses score alike on validation, 0.077 against 0.076, the right-handed one with a tilt of −6.17°. With the tilt held at τ_{IMU} the right-handed azimuth scores 0.145 against 0.079, but that is no test: pointing table v6's tilt takes its sign from v0017's left-handed fit (§5.6), so the held tilt agrees with the left-handed azimuth by construction. The handedness rests on memo 003's `imu_az` test (memo 003, §5.1). The left-handed reruns reproduce the variant (tilt free) and v0018 (tilt held). Memo 003 finds the same zero from the arm ratio. Arm 0's bearing, 172°, lies about 10° from the axis photographed as rx6 (memo 003 leaves that gap open), far nearer it than the other arm, so memo 001's arm labels stand.

5 Beam product v0018

v0018 is v0017 with three inputs changed: pointing table v6 (`v6+4af5ce9`) in place of v5, raster flags v0002 in place of v0001, and the axle tilt held at the IMU's mean (`fit_beam.py --tilt fixed-imu`) instead of fitted. The tooth selection, background, bases, penalty, split rule and every other input are v0017's. The same fit with the tilt free (`--tilt fit`) is kept as the *fitted-tilt variant*, `empirical_raster_v0018/variant_fitted_tilt`.

Table 3: The nested geometry corrections on v0018's samples, fitted with the HFSS beam on the training stripes and scored by the median per-tooth fractional rms on the validation and test stripes; the transmitter is held at memo 002's position. Top: the fitted-tilt variant's candidate sets, with the axle tilt zero or, where listed, fitted. Bottom: v0018's, with the tilt held at τ_{IMU} (the tilt column). "cond." is the condition number of the fit's Jacobian. v0017's geometry is in Table 5.

free	$\delta\alpha$ [deg]	δaz [deg]	δel [deg]	tilt [deg]	cond.	val. / test median rms
fitted-tilt variant: tilt zero, or fitted where listed						
none	+0.00	+0.00	+0.00	+0.00	1.0	0.0903 / 0.0895
polarization	+0.44	+0.00	+0.00	+0.00	1.0	0.0904 / 0.0882
az, el zero	+0.00	−0.02	−0.30	+0.00	1.1	0.0908 / 0.0878
polarization, az, el	+0.46	−0.06	−0.31	+0.00	1.1	0.0907 / 0.0870
axle tilt	+0.00	+0.00	+0.00	+4.74	1.0	0.0758 / 0.0757
az, el zero, tilt	+0.00	−0.03	+0.04	+4.74	1.5	0.0758 / 0.0756
polarization, az, el, tilt	+0.15	−0.04	+0.03	+4.73	1.5	0.0768 / 0.0752
v0018: tilt held at τ_{IMU}						
none	+0.00	+0.00	+0.00	+5.91	1.0	0.0798 / 0.0778
polarization	+0.08	+0.00	+0.00	+5.91	1.0	0.0795 / 0.0776
az, el zero	+0.00	−0.02	+0.10	+5.91	1.1	0.0797 / 0.0775
polarization, az, el	+0.08	−0.03	+0.09	+5.91	1.2	0.0792 / 0.0773

Table 4: The geometry stage on v0018's data with each azimuth handedness, with the axle tilt free (all seven candidate sets of Table 3) and with it held at $\tau_{\mathrm{IMU}} = +5.91°$ (the four sets without the tilt): the coarse scan's best azimuth zero, the bearing of transmitter arm 0 and its rms, the kept corrections and the resulting tilt, and the median per-tooth fractional rms on the validation and test stripes. The right-handed rows negate the table's azimuth, so their validation and test stripes are other samples.

azimuth handedness	axle tilt	az zero [deg]	arm 0 [deg]	coarse rms	corrections kept	tilt [deg]	val. / test
left-hand	free	+16	172	0.119	tilt	+4.74	0.076 / 0.076
left-hand	held	+16	172	0.119	pol., az, el	+5.91	0.079 / 0.077
right-hand	free	−16	84	0.131	pol., az, el, tilt	−6.17	0.077 / 0.075
right-hand	held	−16	84	0.131	polarization	+5.91	0.145 / 0.137

Run on pointing table v5 with v0017's settings, the generator reproduces v0017 to the solver's floor: the exported beam to 0.0021 dB rms on the sampled directions (mean over frequency) and to 0.010 dB within 20° of the boresight. This section's figures, tables and numbers are computed by `beam_v0018.py` from the comparison files of the v0018 build, and Fig. 3's jackknife from the three products' per-sample held-out predictions (Appendix A).

5.1 Pointing table v6 and the recovered rows

Of the raster window's 6,990 rows, 6,817 are `ok` in v6 against 6,788 in v5. v6 corrects the elevation wrap near ±180° on 60 of them (memo 003, §7.1) and makes 31 others suspect, because the box was not quasi-static. On rows `ok` in both, the elevation changes by 0.0032° rms and at most 0.089°. The fit's samples go from 6,552 to 6,579, and those with at least one good fitted tooth from 6,310 to 6,328. Of the 60 recovered rows, 2 lie outside the height era, raster flags v0002 flag 1, and 57 enter the fit. They face the transmitter: in the fit's frame they lie 7.0–9.9° from the boresight (mean 7.5°), so the used samples within 10° of the boresight go from 324 to 384, and within 20° from 723 to 783 (v0018's samples with flags v0001 and v0002).

5.2 Raster flags v0002

Raster flags are built from a reference beam fit, not from the pointing table: `build_raster_flags.py` flags samples whose broadband level k against the reference model departs from the rest, samples that are mostly masked, padding around both, and single-sample tooth spikes. v0001's reference was v0006's PCA fit on pointing

table v2-beta (6,901 samples). The recovered rows are not in it, and the fit counts a sample the flag file does not cover as flagged, so on v0001 none of them would enter. v0002 keeps v0001's 46 teeth and rule parameters and changes only the reference, a PCA fit to the pointing-v6 raster with no flags, in v0018's tilt configuration (6,579 samples); with its defaults the modified builder reproduces v0001 bit for bit. On the 6,518 samples the two share, 222 sample flags are in both, 17 only in v0001 and 28 only in v0002, most of them in the raster's dropout bursts; the scatter of $\ln k$ (1.4826 times its median absolute deviation, over the samples the builder scores) is 0.062 and 0.055. Alone, the change of flags moves v0018's exported beam by 0.013 dB rms on the sampled directions and at most 0.084 dB within 20° of the boresight, and its held-out mean fractional rms from 0.07223 to 0.07217 (better on 42 teeth, worse on 22).

5.3 Geometry

With the tilt held, the geometry stage keeps the polarization and the az and el zeros (Table 5): $\delta\text{az} = -0.030°$, $\delta\text{el} = +0.094°$ and $\delta\alpha = +0.079°$. The HFSS beam fits v0018's validation stripes with a median fractional rms of 0.0792, against 0.0764 for v0017 and 0.0758 for the variant.

Table 5: Geometry of v0017, v0018 and the fitted-tilt variant: the pointing table and raster flags, the coarse scan's azimuth offset and arm-0 angle, the corrections kept, the axle tilt (v0018: the IMU's mean over the kept samples, with memo 003's axis error; the variant: a delete-one-stripe jackknife over the 13 training stripes), the corrections' values (– where not fitted), and the HFSS geometry stage's median per-tooth fractional rms on the validation and test stripes.

	v0017	**v0018**	fitted-tilt variant
pointing table	v5	v6	v6
raster flags	v0001	v0002	v0002
az offset / α_0 [deg]	+16 / 8	+16 / 8	+16 / 8
corrections kept	az, el, tilt	pol., az, el	tilt
tilt τ [deg]	+4.74 (fitted)	$+5.91 \pm 0.15$ (IMU)	$+4.74 \pm 0.23$ (fitted)
δaz zero [deg]	−0.066	−0.030	–
δel zero [deg]	+0.045	+0.094	–
$\delta\alpha$ [deg]	–	+0.079	–
HFSS stage, val. / test	0.0764 / 0.0766	0.0792 / 0.0773	0.0758 / 0.0757

5.4 Held-out prediction

Each product's held-out DPSS beam, fitted on its own training stripes, is scored on v0017's validation samples that all three products share (1,041 of 1,043), with a common good mask, on all 64 fitted teeth (Fig. 3, Table 6). They exclude, by construction, the 57 rows v6 adds (§5.1), so this paired test does not reach them. v0018 scores 0.0722 (mean over teeth of the per-tooth fractional rms; pooled 0.0579) against v0017's 0.0725 (0.0580). Per tooth the change is -2.9×10^{-4} on average and spans -52 to $+28\times10^{-4}$ (the largest increase at 158.2 MHz); v0018 is better on 38 teeth and worse on 26. The worst tooth, the weak arm-0 tooth at 132.8 MHz, goes from 0.1740 to 0.1763. The PCA beams give 0.0742 and 0.0739. The variant scores 0.0724, worse than v0018 on 37 of 64 teeth.

The decision's test. The bottom of Fig. 3 is v0018 minus the fitted-tilt variant per tooth, with a delete-one-stripe jackknife over the 4 validation stripes (1,041 samples) as its error, mean 13×10^{-4} and at most 50×10^{-4}. The difference is positive on 27 teeth and negative on 37, within one error on 36 and within two on 56 of 64; averaged over teeth it is $-2.3 \pm 5.3 \times 10^{-4}$. With 4 stripes the jackknife has 3 degrees of freedom, so its errors are themselves uncertain: 8 teeth lie beyond twice their error (at most 3.9 times it), where a Gaussian would put 3 and a Student t with 3 degrees of freedom 9. Fitting the tilt does not improve held-out prediction, so holding the measured tilt costs nothing there; it does change the beam (§5.6). Against v0017 (grey) the mean change is $-2.9 \pm 5.1 \times 10^{-4}$.

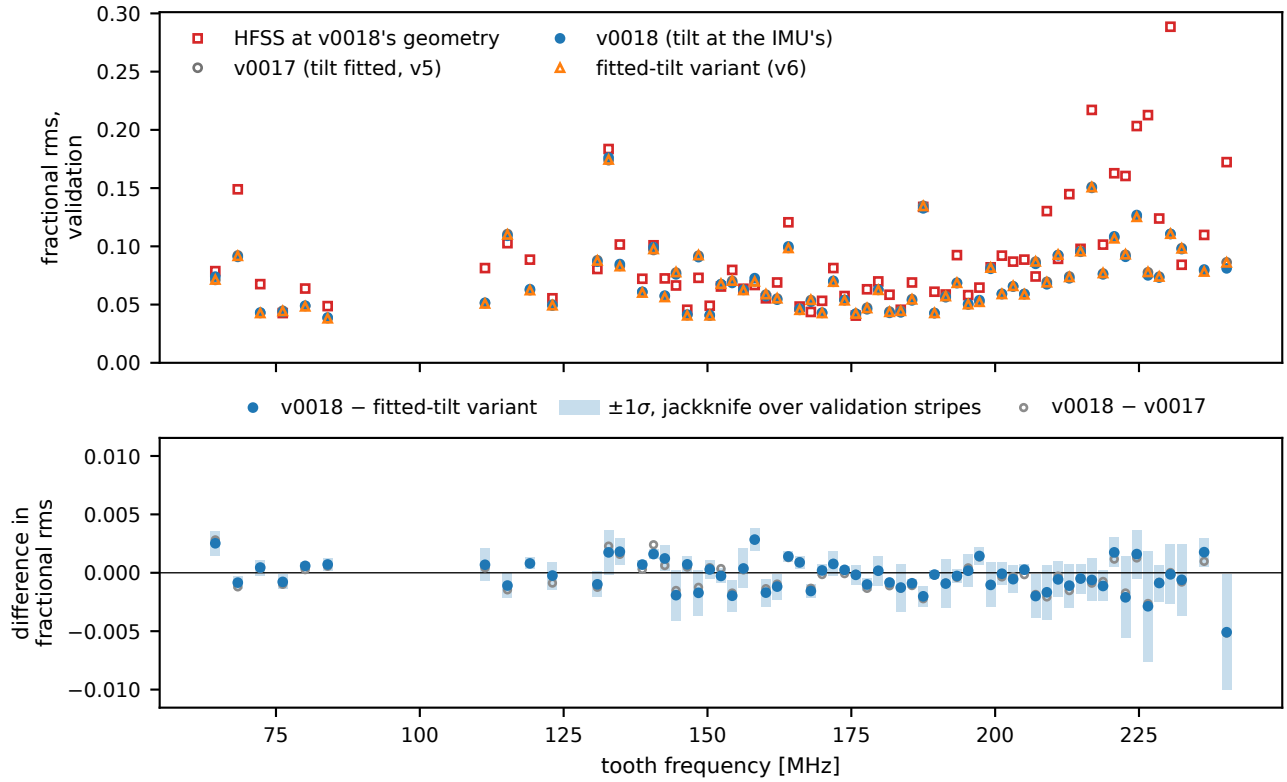


Figure 3: Fitting the tilt does not improve held-out prediction: v0018 holds the measured tilt at no cost to it. Held-out prediction per tooth on v0017's validation stripes (1,041 samples shared by the three products, one good mask). Top: the fractional rms $\|d-m\|/\|d\|$ of each product's held-out DPSS beam, fitted on its training stripes, and of HFSS with v0018's geometry and one gain per tooth. Bottom: v0018 minus the fitted-tilt variant (blue), with its $\pm1\sigma$ error from a delete-one-stripe jackknife over the 4 validation stripes (shaded), and v0018 minus v0017 (grey), mostly hidden under the blue points. Positive: v0018 predicts worse. The v0018−variant difference lies within 2σ on 56 of 64 teeth (the jackknife has 3 degrees of freedom) and averages $-2.3\pm5.3\times10^{-4}$.

5.5 The beam's change

Each exported DPSS map's total power, per frequency over its peak, is compared with v0017's (Fig. 4, Table 7). On the sampled pixels (42% of the sphere, those v0017's export counts as visited; v0018's counts 41%) the rms of the dB ratio is weighted by v0017's visits times its power. v0018's beam moves by 0.074 dB rms (mean over frequency; worst 0.132 dB at 240.2 MHz) and by at most 0.28 dB within 20° of the boresight (at 240.2 MHz), where its mean change is +0.0034 dB. The variant, which keeps v0017's tilt, moves by 0.014 dB, and the flags alone by 0.013 dB: the tilt moves v0018's beam, which takes up the 1.18° between the two tilts (§5.6). The PCA beam moves by 0.070 dB. Where the raster never looked, the peak below 100 MHz relative to HFSS's is 1.068 (at 64.5 MHz) against v0017's 1.057 and the variant's 1.047; v0016's low-band lobe (Table 2) does not return. That ratio scales each map to unit visit-weighted mean on its own sampled pixels; it is not Table 2's peak ratio. At the top of the band the unsampled peak of all three products rises to 1.24 at 240.2 MHz (Fig. 4c), and above 90 MHz v0018's worst peak ratio in Table 2 exceeds v0016's. That is where only arm 1 is fitted (§2), and the raster does not constrain it.

5.6 The axle tilt

Why v0017's tilt differs from the IMU's. v0017's tilt was a fitted constant. Pointing table v5 carried no tilt along the axle, so the geometry stage fitted one, +4.74°, to the HFSS beam. The IMU reads +5.91° over the same raster (Fig. 5). Pointing table v6's tilt takes its sign from v0017's fitted +4.74° (memo 003, §7.2: the sensor's convention is not confirmed), and only its sign; its size is the IMU's.

Table 6: Held-out fractional rms on v0017's validation samples (Fig. 3): the mean over the 64 teeth, the worst tooth and the value pooled over teeth, for the DPSS and PCA held-out beams; below, the per-tooth DPSS differences in units of 10^{-4}, their mean and range, and the number of teeth on which the first product is better or worse.

	DPSS			PCA	
product	mean	worst (132.8 MHz)	pooled	mean	pooled
v0017	0.0725	0.1740	0.0580	0.0742	0.0590
v0018	0.0722	0.1763	0.0579	0.0739	0.0583
fitted-tilt variant	0.0724	0.1745	0.0578	0.0742	0.0588
difference (DPSS), 10^{-4}	mean	range	better / worse		
v0018 − v0017	−2.9	−52 . . . + 28	38 / 26		
variant − v0017	−0.6	−8 . . . + 8	37 / 27		
variant − v0018	+2.3	−28 . . . + 51	27 / 37		

Table 7: The exported DPSS beam's change in dB (Fig. 4): v0018 and the variant against v0017, the solver's floor (the generator on pointing table v5 against v0017), and v0018 on flags v0002 against the same fit on flags v0001. "Unsampled" is over pixels the raster never visited.

DPSS, dB	**v0018**	variant	solver floor	flags v0002 alone
rms on sampled pixels: mean over f / worst	0.074 / 0.132	0.014 / 0.053	0.002 / 0.007	0.013 / 0.049
max $\lvert\Delta\rvert$ on a sampled pixel	2.74	1.09	0.27	0.93
mean within 20°: mean over f	+0.0034	+0.0011	−0.0003	+0.0028
max $\lvert\Delta\rvert$ within 20°: worst f	0.276	0.080	0.010	0.084
max $\lvert\Delta\rvert$ unsampled	5.37	1.00	0.32	1.11

The fitted tilt. Fitted on pointing table v6 and flags v0002, the tilt comes out $+4.74 \pm 0.23°$, the error a delete-one-stripe jackknife over the 13 training stripes, whose replicates span +4.56 to +4.82°; each geometry fit starts from zero tilt (`tx_fit.fit_geometry`). The variant keeps the tilt alone; with the az and el zeros added it ties on validation (0.07584 against 0.07583), so that choice is not an effect of the recovered rows.

The raster cannot separate the tilt from the beam's shape. Only the geometry stage, which uses the HFSS beam, prefers the fitted tilt, by 4.3% in validation median rms (0.0758 against 0.0792). Once the empirical beam is fitted, the variant predicts the held-out stripes no better than v0018 (0.0724 against 0.0722, worse on 37 of 64 teeth; Fig. 3). v0018's beam takes up the difference as a change of 0.074 dB rms, and at most 0.28 dB near the boresight (Fig. 4). The two fits also differ in more than the tilt: the variant holds the azimuth, elevation and polarization corrections at zero, and v0018 fits them with the tilt held. HFSS's shape trades against the tilt, so the raster cannot separate a change of tilt from a small change of the beam's shape, and the fitted value says nothing about the mount.

v0018 holds the IMU's tilt. The antenna is bolted to the mount on the box, so the axle tilt the beam sees is the IMU's. On the 07-17 night a sky fit with no free sky field is consistent with the IMU's tilt (memo 003, §7). The fitted-tilt variant is kept as the documented alternative for users: v0018's beam moves from v0017's by 0.074 dB rms on the sampled pixels and at most 0.28 dB within 20° of the boresight, the variant's by 0.014 dB (§5.5).

The raster's tilt dip. The tilt dips during the raster, 20:54–21:01 UTC (memo 003, §7). Over the fit's kept samples the IMU reads +4.77° in the window (761 samples) and +6.07° outside it (5,567). On the window's 260 held-out samples of the validation and test stripes, v0018, with one tilt for the raster, predicts as well as the variant: 0.0627 against 0.0626 (mean over teeth of the per-tooth fractional rms); outside it the variant is the better, 0.0763 against 0.0770. Both, and HFSS (0.0798 against 0.0959), score better in the window than outside, so that difference is in the directions sampled, not the tilt. Refitting the variant's tilt alone with the HFSS beam gives +4.74° on all samples, +4.88° without the window and +4.05° on the window's 501 training

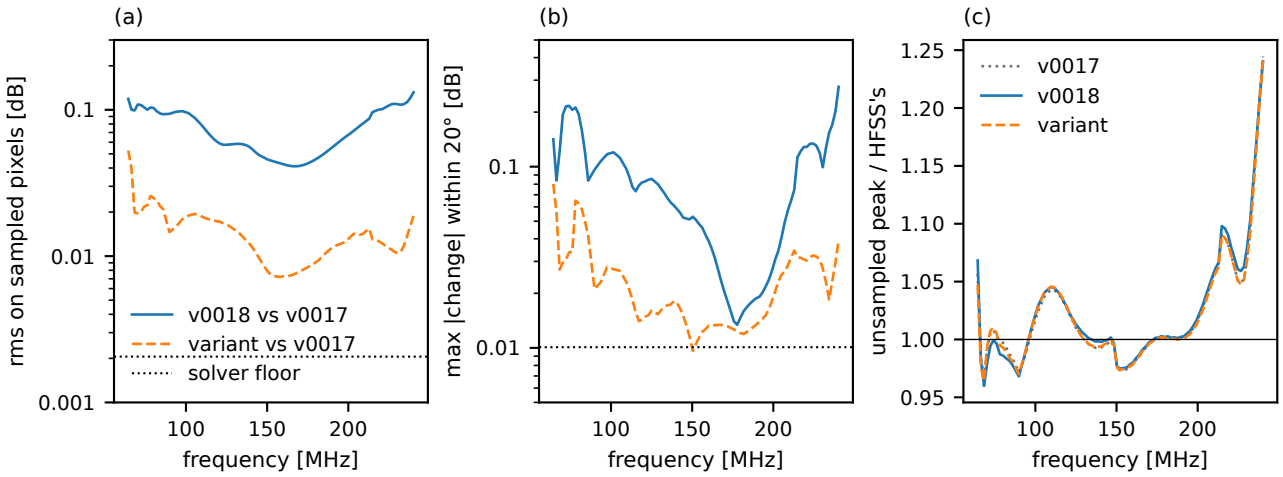


Figure 4: v0018's beam moves several times more than the variant's, which keeps v0017's tilt: the exported DPSS beam's change from v0017 against frequency, for v0018 (solid) and the fitted-tilt variant (dashed). (a) The rms of the dB ratio of total power on the sampled pixels, weighted by v0017's visits times its power. (b) The largest absolute change in dB within 20° of the boresight. Dotted: the solver's floor, the generator on pointing table v5 against v0017 ((a) its mean over frequency, (b) its worst). (c) Each map's peak off the sampled pixels over HFSS's, each map scaled to unit visit-weighted mean on its own sampled pixels; the three curves nearly coincide.

samples alone (v0017's corrections: +4.91° and +3.41°), with no error computed: consistent with a radio tilt that follows the dip by less than the IMU's.

6 Results

The figures and scores of this section, and the raster maps of Appendix B, are computed from v0018's per-sample products.

Figure 6 shows the data and both models for two teeth. Both follow the large modulation as the boresight passes the transmitter once per revolution; the differences are in the shoulders and the sidelobes.

Raster maps. The direct comparison is the raster itself (Fig. 7): each tooth's background-subtracted power at every good sample, placed at its azimuth and elevation in the product's fitted geometry (table azimuth plus the +16° offset and δaz, IMU elevation plus δel; the axle tilt enters the model but not these coordinates), beside the HFSS prediction at the same samples with that tooth's gain fitted on the training stripes, and their difference. Appendix B shows all 101 candidate teeth, 50.8–246.1 MHz, one by one, including the 37 the selection dropped (Fig. 1); for those the HFSS gain is fitted the same way and nothing else is. The product holds only the selected teeth, so the excluded ones are rebuilt from the raster with the product's own generator (`fit_beam.load_data`: same samples, background and masks), which reproduces the selected teeth to <0.001% rms of each tooth.

Table 8: Per-tooth fractional rms $\|d-m\|/\|d\|$ over the 64 teeth: the mean weighted by good-sample count, the 90th percentile and the worst tooth. Only the held-out fits' test column is independent of every fitted choice; HFSS has one gain per tooth fitted on the training stripes.

model	train	validation	test
HFSS	0.093 / 0.148 / 0.266	0.092 / 0.161 / 0.287	0.094 / 0.156 / 0.276
DPSS held-out	0.070 / 0.099 / 0.162	0.072 / 0.106 / 0.175	0.077 / 0.103 / 0.246
PCA held-out	0.072 / 0.107 / 0.168	0.073 / 0.114 / 0.182	0.079 / 0.115 / 0.248
DPSS refit	0.070 / 0.099 / 0.164	0.069 / 0.103 / 0.160	0.073 / 0.101 / 0.238
DPSS exported	0.070 / 0.099 / 0.164	0.069 / 0.103 / 0.160	0.073 / 0.101 / 0.238

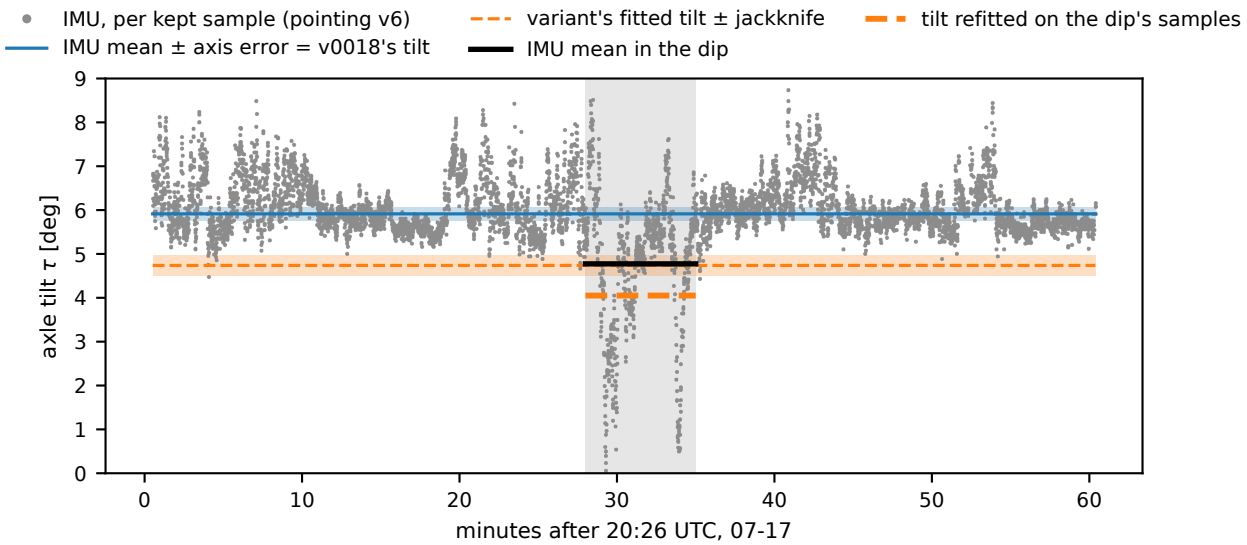


Figure 5: The IMU's tilt lies above the variant's fitted tilt through the raster and falls during the 20:54–21:01 UTC dip. The elevation axle's tilt over the 07-17 raster: pointing table v6's tilt per sample on v0018's kept samples (grey; samples with at least one good fitted tooth), their mean, which is v0018's tilt, with memo 003's axis error (blue), and the fitted-tilt variant's tilt with its jackknife error (orange; v0017's fitted +4.74° coincides with it). The grey band is the 20:54–21:01 UTC dip: the IMU's mean over it (black) and the tilt refitted with the HFSS beam on its samples alone (orange, thick dashed).

Scores. On the test stripes the held-out DPSS beam reaches 0.077 (90th percentile 0.103, worst 0.246 at 148.4 MHz) against HFSS's 0.094 (0.156, 0.276 at 230.5 MHz) (Table 8). PCA does the same (0.079). The empirical beam is better on 45 of 64 teeth (Fig. 8). The worst empirical tooth, 148.4 MHz, is one of the weak arm-0 teeth, whose background matters most. v0017 scored 0.076 (DPSS), 0.078 (PCA) and 0.091 (HFSS) on its own test stripes (1,194 samples; better than HFSS on 44 of 64 teeth). The stripes are the same azimuth bins, but the samples and flags differ (§5.1), so the paired comparison of the two products is §5's, on the validation samples they share.

Table 9: Test-stripe fractional rms by band (mean over teeth weighted by sample count), and the white-noise floor.

band [MHz]	teeth	HFSS	DPSS held-out	PCA held-out	noise floor
50–87	6	0.083	0.066	0.075	0.014
87–145	11	0.086	0.082	0.085	0.015
145–190	23	0.077	0.073	0.073	0.012
190–235	22	0.115	0.082	0.084	0.014
235–245	2	0.116	0.073	0.070	0.014

The residual is not noise. The white-noise part of each tooth's residual, estimated from differences of successive samples, is 0.014 of the tooth's rms on average (worst 0.035, the weak 132.8 MHz arm-0 tooth). The test residual is ∼6× larger, so it is structure that changes slowly along the raster: beam features the model does not capture, background errors, or geometry. Binned by the transmitter's angle from the boresight (Fig. 9), the fractional residual is smallest near the boresight and near the back of the antenna, where most of the tooth power is, and largest at 60–140°, where the power is one to two orders of magnitude lower. The empirical beam improves on HFSS at every angle.

The beam. Figure 10 compares the exported DPSS beam with HFSS at three frequencies. Weighting each sampled pixel by its number of visits and by the beam's power there, the two differ by 0.1–0.3 dB rms, and at

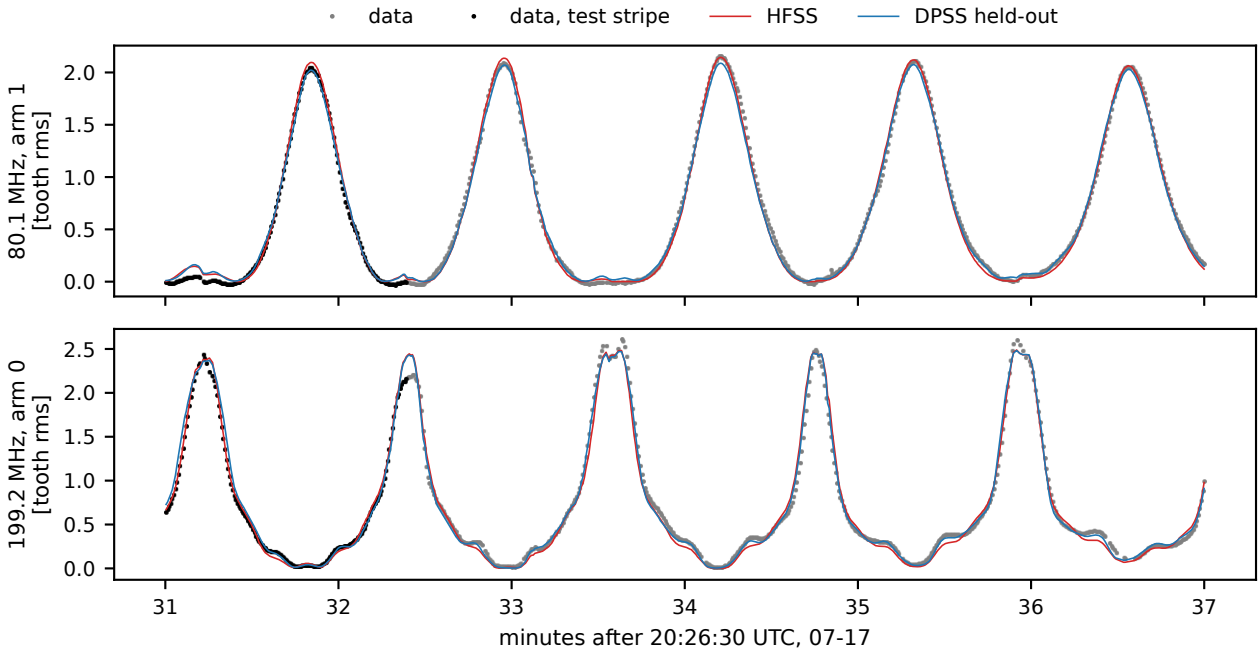


Figure 6: Both models follow the modulation as the boresight passes the transmitter; they differ in the shoulders and sidelobes. Two teeth over six minutes of the raster, in units of each tooth's rms: data (grey; black on a test stripe), the HFSS beam with its fitted gain (red), and the held-out DPSS beam (blue). Each peak is one elevation revolution passing the transmitter. The time axis counts from the first fit sample (Fig. 5 counts from 20:26:00), and the six minutes begin inside the 20:54–21:01 UTC tilt dip (§5.6). The residual is in the raster maps (Fig. 7).

least 98% of that weight is within 1 dB; single low-power pixels differ by up to 3 dB. The raster samples 41% of the HEALPix pixels; elsewhere the beam is set by the HFSS prior and the smoothness of the basis, and its differences from HFSS there are not measurements.

7 The HFSS beam solved at 1 MHz

The HFSS beam has 52 frequencies, 3.90625 MHz apart, and every use of it between them interpolates in frequency: the beam fitter linearly (§3), other analyses with four-node (cubic) Lagrange weights. The same HFSS model (same project, design, port and excitation) has now been solved at every integer frequency from 50 to 250 MHz, 201 frequencies, on the same 1° far-field grid. At 125 MHz, where both grids have a solve, the two agree to 8×10^{-7} in field (relative rms): they are the same model.

The 52-frequency beam was already smooth enough in frequency. At each integer frequency from 51 to 249 MHz, inside the coarse grid so nothing is extrapolated, the 52-frequency fields are interpolated component by component in Cartesian coordinates and compared with the 1 MHz solve, both scaled to unit mean power over the sphere and made symmetric about the antenna's x mirror plane (Fig. 11). Linear interpolation, as in the beam fitter, reproduces the 1 MHz beam to at most 0.28% of the peak power in any direction (at 236 MHz; 0.08% on average over frequency), to 0.41% in field (relative rms, at 53 MHz) and to 0.19% in power (relative rms, at 232 MHz). The beam fit's held-out residual, the same relative rms of power on the raster's samples, is at least 3.9% at every scored frequency (§8). Four-node interpolation does 26 times better: at most 1.1×10^{-4} of the peak power (248 MHz) and 1.7×10^{-4} in field (52 MHz, where the stencil is one-sided). The error is smallest near the coarse solve frequencies and largest between them. Interpolating the 52-frequency beam therefore cost nothing this memo can measure, and the 1 MHz product removes frequency interpolation as a question for every later use.

Night fit. In a night fit like memo 008's, at 18 native channels (69.1–71.5, 147.2–149.7, 205.8–208.3 MHz) over 04:12–11:25 UTC on 07-17, replacing the interpolated 52-frequency beam with the 1 MHz beam changes the held-out mean squared error by +2.3 parts per million (95% interval 0.9–4.0, from a bootstrap over 7 held-out blocks).

The product. `derived/beam/hfss_bowtie_1mhz_v0001/` holds the 201-frequency beam in two forms. `bowtie_beam_1mhz.npz` is in the HFSS file format (HEALPix nside 32; read by `read_beam` and `HFSSBeamSet`), normalized as the empirical exports are: radial part removed and unit mean power over the sphere at each frequency. Its maps are synthesized at the pixel centres from the exact spherical-harmonic transform of the 1° grid, to $\ell = 24$, above which the field holds 1×10^{-12} of its power; there is no angular resampling. At 125 and 250 MHz, the frequencies it shares with `bowtie_beam.npz`, the two differ by 0.17% and 0.35% in field rms; the older file was resampled onto HEALPix from the HFSS grid. `bowtie_1MHz_mwss_v0001.npz` is the native 1° grid: the complex Cartesian field $r\mathbf{E}$ in mV for 1 W at the port, with the port's reflection coefficient. Neither file is made symmetric; the solver's asymmetry about the x mirror plane is at most 0.048% of the field (250 MHz). The directory's README gives the provenance and the two scripts that made the HEALPix file and Fig. 11. Where the simulated port sits relative to the built antenna's balun and coax, and the model's absolute radiated power, are not established, so neither file sets an absolute efficiency; that does not affect the normalized pattern compared here.

8 Caveats

- **Phase and circular polarization** are not constrained: the data are powers. The coefficients are regularized toward HFSS, which fixes what the data cannot.
- **The test is a held-out stripe of azimuth, not a fresh experiment.** The stripes share the session, the background model and the geometry with the training data.
- **Penalty**: the DPSS penalty is chosen by the exported beam's agreement with HFSS in total power (Table 2), not by a held-out score, which measures only the power in the transmitter's polarization along the raster's track. 10^{-2} is the strongest penalty tried, so the beam where the data are silent is mostly HFSS's. Memo 008's low-band numbers use the exported beam.
- **Background**: the transmitter-off control test of the background (about 1% bias, 2–3% scatter) was made with the earlier 15-mode background and has not been rerun for the 21-mode background now used; it matters most for the weak arm-0 teeth. Raster flags v0002 were built on a reference fit to the same raster in v0018's configuration (§5.2).
- **Geometry**: the axle tilt is held at the IMU's mean over the raster (the antenna is bolted to the mount on the box), with the transmitter held at memo 002's position (§5.6). A fitted tilt trades against the beam's shape and says nothing about the mount; the fitted-tilt variant is the documented alternative (§5.5). One tilt serves the whole raster, though the IMU's falls to +4.77° over 20:54–21:01 UTC; the held-out residual does not grow there. The tilt's sign is taken from the raster beam fit (memo 003, §7.2). The fitted elevation zero (+0.094°) does not match the IMU offset implied by the LIDAR (memo 003, §6).
- **Elevation error.** Pointing table v6's elevation may carry a twice-per-turn error of 0.55° (possibly up to ≈0.9°), because box-air's IMU axes across the axle are 91.08° apart (memo 003, §7). An error of 0.55° changes this beam by 1.2–2.3 % of its peak over its −10 dB region (64–240 MHz), below the fit's held-out residual at every scored frequency (at least 3.9 %). It is not corrected in this release.
- **One polarization at the band's edges.** Below 132.8 and above 230.5 MHz only arm-1 teeth are fitted (§2). The upper band is where v0018's beam changes most from v0017's (0.132 dB rms at 240.2 MHz; at most 0.28 dB within 20° of the boresight, at 240.2 MHz) and where Fig. 3's jackknife error is largest (50×10^{-4} at 240.2 MHz).
- The DPSS fits at $\lambda = 10^{-4}$, v0018's and the variant's, stopped at the 3000-iteration limit; the PCA fits at 10^{-4} converged after 2770 and 2457 iterations, and the chosen fits converged (each product's `optimization.json`).

9 Open questions

- What the remaining structure is: a higher $\ell_{\max}$ or a richer spectral basis where the residual is largest, against background and geometry errors.
- Which features of the HFSS beam's shape trade against the axle tilt, so that the HFSS-only geometry stage prefers +4.74° to the IMU's +5.91° (§5.6).
- The gap of about 10° between the polarization's arm 0 and the photographed rx6 axis (memo 003, open questions).
- A transmitter-off control for the 21-mode background.

Revision history

2026-11-09 The HFSS model solved at 1 MHz spacing (§7): interpolating the 52-frequency beam in frequency reproduces it to at most 0.28% of the peak power (linear) and 1.1×10^{-4} (four-node), so the 52-frequency beam was adequate. The 201-frequency beam is released as `derived/beam/hfss_bowtie_1mhz_v0001`.

2026-11-06 Memo 003's IMU tilt now comes from readings with the accelerometer's bias removed; the numbers this memo takes from memo 003 do not change. A caveat gives the twice-per-turn elevation error that memo 003 now finds in pointing table v6 and its change to this beam, below the held-out residual (§8).

2026-11-04 Moved to beam product v0018 (§5): pointing table v6, raster flags v0002, and the elevation axle's tilt held at the box IMU's mean over the raster, +5.91°, instead of fitted. v0017's fitted +4.74° was a constant fitted with the HFSS beam, before the pointing table carried the tilt. Fitted again on v6 it stays near +4.74° but predicts the held-out stripes no better than the IMU's tilt, so the raster cannot separate a change of tilt from a small change of the beam's shape; the antenna is bolted to the mount on the box, so v0018 takes the tilt from the IMU, and the fitted-tilt variant is the documented alternative. The 2026-10-31 entry's "The IMU measures the tilt's sign" is superseded: the sign is taken from the raster beam fit (memo 003, §7.2). v6's elevation fix adds 57 samples within 10° of the boresight, which flags v0002 cover. v0018's beam differs from v0017's by 0.074 dB rms on the sampled directions, and its held-out fractional rms by $-2.9 \pm 5.1 \times 10^{-4}$ on average; against the fitted-tilt variant by $-2.3 \pm 5.3 \times 10^{-4}$ (Fig. 3, with a jackknife over the validation stripes). Every section now draws on v0018: the scores, maps, residual by angle, raster maps, geometry tables and the handedness check; the summary compares v0018 with v0017, and the comparison with v0013 is replaced by the table of products (Table 10).

2026-10-31 Moved to beam product v0017: pointing table v5 (azimuth left-handed about the boresight, memo 003), memo 002's `v0005` posterior-mean positions, and a tilt of the elevation axle in place of the transmitter offsets, which had run to their bound. The IMU measures the tilt's sign. The DPSS penalty is held at 10^{-2}, because the weaker penalty the validation scores prefer (v0016) puts spurious lobes in the exported beam at the bottom of the band. The comparison is now with v0013.

2026-10-08 Moved to beam product v0013, the same fit on geometry release `v0004`; adds the comparison with v0012's `v0001` geometry.

2026-10-02 First version, on beam product v0012. Adds count-weighted scores with worst cases, the white-noise floor, the residual by angle, and the handedness check of the geometry stage.

A Reproducing this memo

The beam product is regenerated, from the eigsep checkout root, by `python data-analysis/scripts/marjum-2026-07/fit_beam.py --output marjum-2026-07/derived/beam/empirical_raster_v0018 --penalty dpss=1e-2`, and otherwise the defaults of `fit_beam.py` on data-analysis `main` (`912ae15`): pointing table v6, geometry release `v0005` with memo 002's posterior-mean positions, the axle tilt held at the IMU's (`--tilt fixed-imu`) and the other geometry candidates, teeth `tooth_selection_v0002.json`, flags `raster_flags_v0002.npz`, DPSS and PCA bases, 3000 iterations, background concentration 10^{-6}. It ran at `e05675a`, the branch commit that `912ae15` merged unchanged, with `eigsep_data` at `9ac9b93`. The variant adds `--tilt fit` and goes in `variant_fitted_tilt/`. Table 10 lists the products this memo has

used. This memo's figures, tables and numbers come from `python make_products.py` in this directory (set `EIGSEP_CAMPAIGN_ROOT` to the campaign directory that holds `derived/beam/` when running from a worktree), then `pdflatex memo-005.tex` twice; set `EIGSEP_DATA_ANALYSIS` to use a data-analysis checkout other than the one in this checkout. The script checks the SHA-256 of every gitignored input. It also writes `tables/v0018_heldout_jackknife.csv`, the per-tooth held-out scores of v0017, v0018 and the variant on their shared validation samples with the jackknife of Fig. 3, and checks the scores against the v0018 build's. The axle-tilt numbers quoted from memo 003 (§7) are read from its `tables/summary.json` (SHA-256 below); `python make_products.py --only tilt` merges them alone. `python make_products.py --only v0018` rebuilds §5 alone, with those tilt numbers, by running `beam_v0018.py` on small files of the v0018 build copied into `inputs/v0018/` (SHA-256 checked; below) and on the jackknife table; `python beam_v0018.py --extract <product> <pointing table>` writes the per-sample tilt file from the product and pointing table v6. `python make_products.py --only hfss1mhz` rebuilds §7 alone, by running `beam_hfss_1mhz.py` on three small files in `inputs/hfss_1mhz/` (SHA-256 checked; below): the interpolation errors of Fig. 11, written by `compare_frequency_interpolation.py` in the product's directory from the two native exports; the product's provenance and checks, written by its `make_bowtie_beam_1mhz.py`; and the night fit's held-out scores with each beam.

Table 10: The beam products this memo has used, from each product's provenance: the pointing table (v2.0's azimuth is the right-handed mirror of v5's and v6's), the antenna and transmitter positions (geometry release; "mean": memo 002's posterior mean, else the release state), the geometry corrections kept, the axle tilt, the raster flags and the DPSS penalty. v0016 is v0017's fit at the penalty its selection rule chose (Table 2).

product	pointing	positions	corrections kept	tilt [deg]	flags	DPSS λ
v0012	v2.0	v0001	pol., az, el, tx at bound	–	v0001	10^{-2}
v0013	v2.0	v0004	pol., az, el, tx at bound	–	v0001	10^{-2}
v0016	v5	v0005 mean	az, el, tilt	+4.74 (fitted)	v0001	10^{-3}
v0017	v5	v0005 mean	az, el, tilt	+4.74 (fitted)	v0001	10^{-2}
v0018	v6	v0005 mean	pol., az, el	+5.91 (IMU)	v0002	10^{-2}
v0018 variant	v6	v0005 mean	tilt	+4.74 (fitted)	v0002	10^{-2}

input	SHA-256 (first 16)
derived/beam/empirical_raster_v0018/provenance.json	1dcbacb2127b8fc2
derived/beam/empirical_raster_v0018/optimization.json	1b0d3df98f777609
derived/beam/empirical_raster_v0018/metrics.csv	26e84f8fddaba4fc
derived/beam/empirical_raster_v0018/geometry_comparison.csv	b25fb38ef6741227
derived/beam/empirical_raster_v0018/az_offset_alpha_scan.csv	21df1975f9dc4196
derived/beam/empirical_raster_v0018/spatial_order.csv	883f6dec2fbfd171
derived/beam/empirical_raster_v0018/dpss/diagnostics.npz	d43fe9ff25932bf1
derived/beam/empirical_raster_v0018/dpss/empirical_beam.npz	10443529c81f83c8
derived/beam/empirical_raster_v0018/dpss/basis_selection.csv	a0ffd68d13d372aa
derived/beam/empirical_raster_v0018/pca/diagnostics.npz	cb2eb1d02a57926a
derived/beam/empirical_raster_v0018/pca/basis_selection.csv	f254c86fe09d1c2a
derived/beam/empirical_raster_v0018/variant_fitted_tilt/provenance.json	d0fd02d40efa9258
derived/beam/empirical_raster_v0018/variant_fitted_tilt/geometry_comparison.csv	de41c71821b9b665
derived/beam/empirical_raster_v0018/variant_fitted_tilt/dpss/diagnostics.npz	3e9d512530a32b2b
derived/beam/empirical_raster_v0017/provenance.json	ca77de003dfa1206
derived/beam/empirical_raster_v0017/dpss/diagnostics.npz	d3f0dd9e3abc7a38
derived/beam/empirical_raster_v0017/pca/diagnostics.npz	32192886739b3f39
derived/beam/empirical_raster_v0013/provenance.json	37b9f6b2432200c3
derived/beam/empirical_raster_v0012/provenance.json	d1221e804109cde4
derived/beam/empirical_raster_v0016/provenance.json	a2956096ed491d50
derived/beam/empirical_raster_v0016/dpss/empirical_beam.npz	57b943bbb8469b8a
derived/beam/raster_flags_v0002.npz	af12045007d9557b
derived/beam/tooth_selection_v0002.csv	acd4e893bc6c1899
derived/beam/tooth_selection_v0002.json	08e4d5e3e48ee9e5
eigsep_data/hfss_beam_maps/bowtie_beam.npz	c4046f5e9332cd57
memos/memo-002-marjum-2026-07-geometry/tables/summary.json	7b1f0684903e71c5
beam product generated by data-analysis/scripts/marjum-2026-07/fit_beam.py	
data-analysis	e05675a
eigsep_data	9ac9b93
eigsep_base	840c10b
eigsep	not recorded
eigsep (memo) commit	fb70e1f
eigsep_data commit (HFSS file, geometry model)	9ac9b93
data-analysis commit (raster rebuild, fit_beam.py)	e05675a
generated	2026-11-05 09:08 UTC

axle-tilt numbers (Geometry)	
memos/memo-003-marjum-2026-07-pointing/tables/summary.json	cd5e65b794e9a3df
eigsep (memo) commit, tilt numbers	caf7653

beam v0018 input (§5)	SHA-256 (first 16)
inputs/v0018/beam_change_dpss.csv	a5b4cffb61fb0cda
inputs/v0018/compare_v0018.json	8f797c63290080ea
inputs/v0018/diff_flags.json	c4afd5a0833d8d21
inputs/v0018/flags_effect.json	6b91c0e4ba486689
inputs/v0018/heldout_validation_dpss.csv	108d9199286d45a9
inputs/v0018/optimization_v0018.json	1b0d3df98f777609
inputs/v0018/optimization_variant.json	4d61d316a43eaacb
inputs/v0018/provenance_v0018.json	1dcbacb2127b8fc2
inputs/v0018/regression_floor.json	9adf6a517da3497f
inputs/v0018/samples_v0018.npz	3088a620aad0f617
inputs/v0018/tilt_jackknife.csv	b7dd96ce4d2f90b7
inputs/v0018/window_check.json	d6a341d84c9fe622
pointing table v6 (v6+4af5ce9)	cb657759fc29c442
raster flags v0002	af12045007d9557b
v0018 generated by fit_beam.py: data-analysis / eigsep_data / eigsep_base	e05675a / 9ac9b93 / 840c10b
eigsep (memo) commit, v0018 products	fb70e1f

1 MHz HFSS beam (§7)	SHA-256
raw HFSS export, bowtie_1MHz_complex.npz	827c11c0254ad405
native grid, bowtie_1MHz_mwss_v0001.npz	9b40552469364304
HFSS format, bowtie_beam_1mhz.npz	2d56177743ab2863
52-frequency native export, bowtie_native_mwss_L180.npz	8c63a213cbdd77c1
inputs/hfss_1mhz/interpolation_error.json	9ae5702ad5dba188
inputs/hfss_1mhz/provenance_hfss_bowtie_1mhz_v0001.json	dc67c6d1e564954d
inputs/hfss_1mhz/nightfit_beam_swap.json	58903800c96e6f91
eigsep (memo) commit, 1 MHz section	b930578-dirty

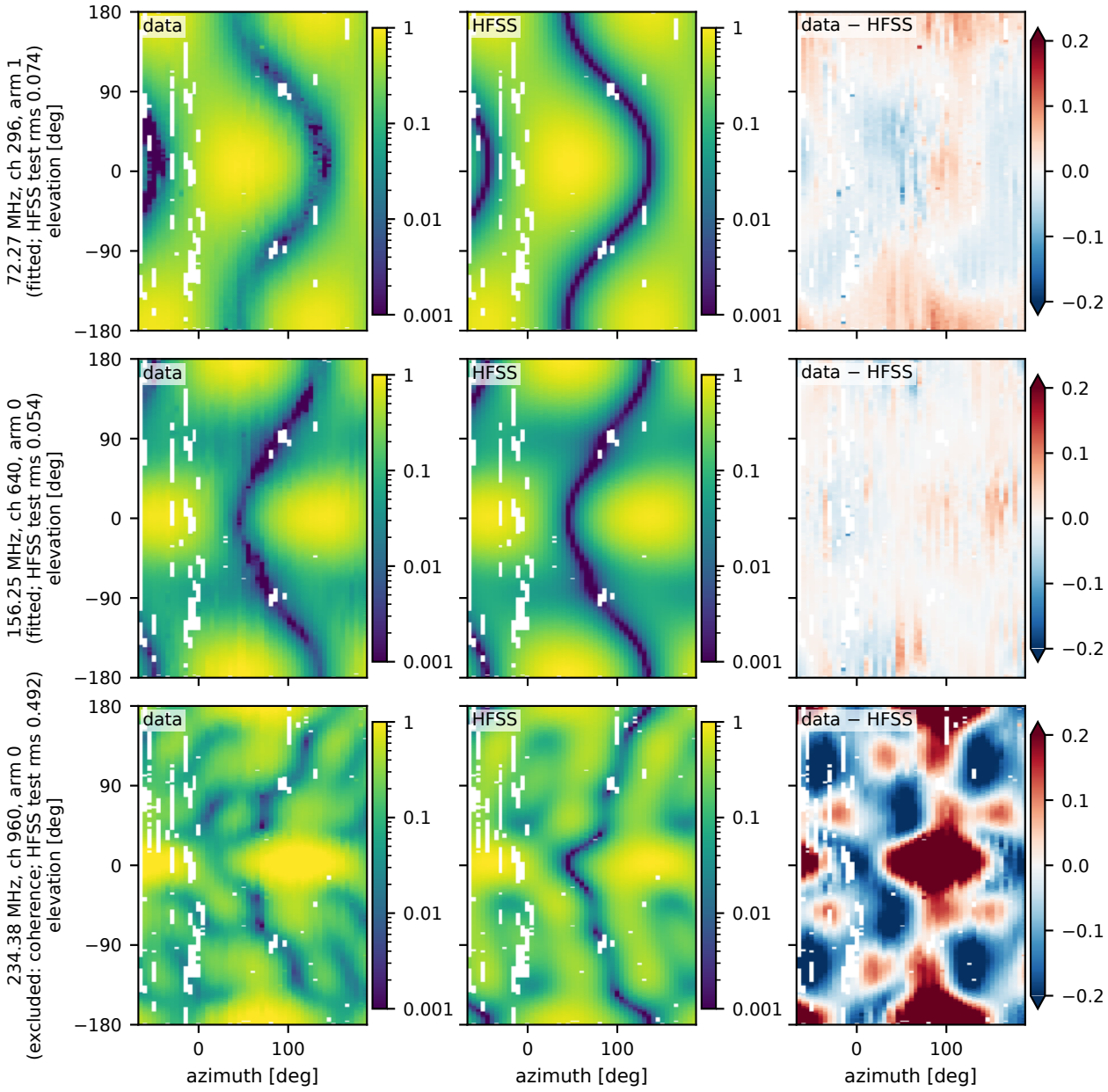


Figure 7: Raster maps of three teeth on azimuth (from the highline, fitted geometry, left-handed about the boresight) and elevation (0 at zenith, ±180° at nadir): data (tooth minus background), the HFSS model with one fitted gain, and data minus model. Every panel is divided by the HFSS model's peak over the raster (its fitted gain times its peak coupling), so all teeth share fixed scales: data and model on a log scale from 10^{-3} to 1 (lower values are drawn at the floor), the residual from −0.2 to +0.2 of the peak. A good fit leaves a pale residual panel; a poor one saturates. Only good samples are drawn. The labels give each tooth's HFSS test-stripe fractional rms; 234.38 MHz was excluded from the fit by the coherence test.

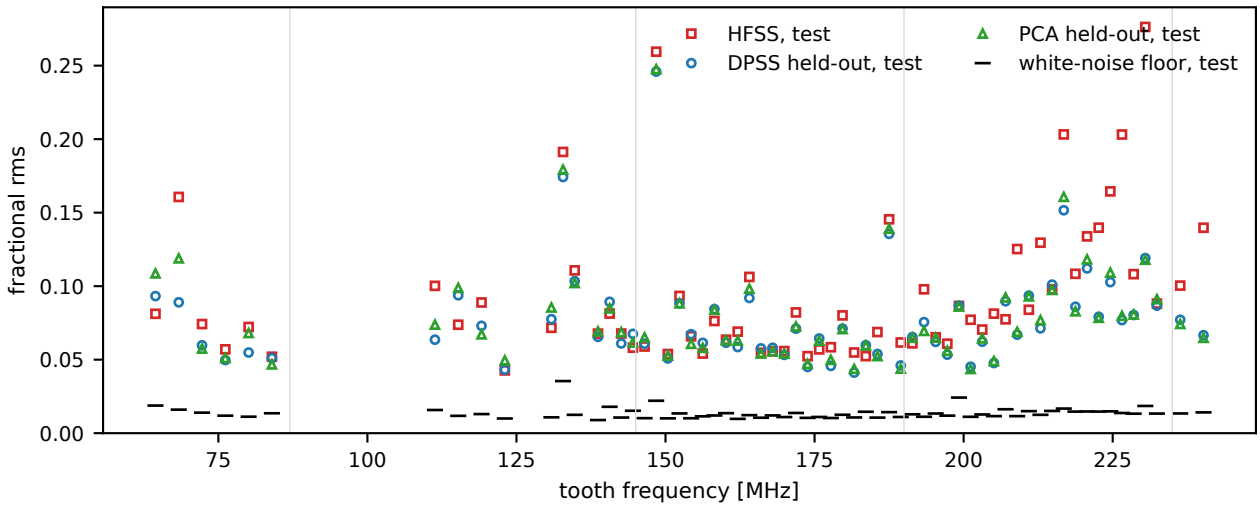


Figure 8: The held-out empirical beams beat HFSS on 45 of 64 teeth. Per-tooth fractional rms on the test stripes for HFSS and the two held-out empirical beams, with the white-noise floor of each tooth's residual. Thin lines mark the band edges of Table 9.

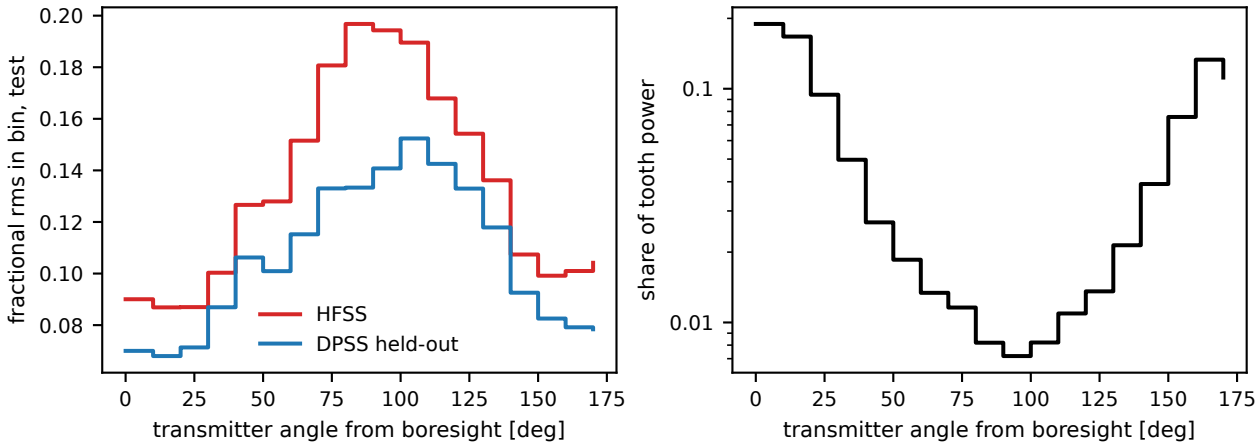


Figure 9: The fractional residual is largest 60–140° from the boresight, where the tooth power is lowest, and the empirical beam improves on HFSS at every angle. Left: fractional rms on the test stripes, pooled over teeth (each tooth scaled to unit mean power), in 10° bins of the transmitter's angle from the boresight. Right: the share of tooth power in each bin. The antenna has no ground plane, so it receives about as much from behind as from in front.

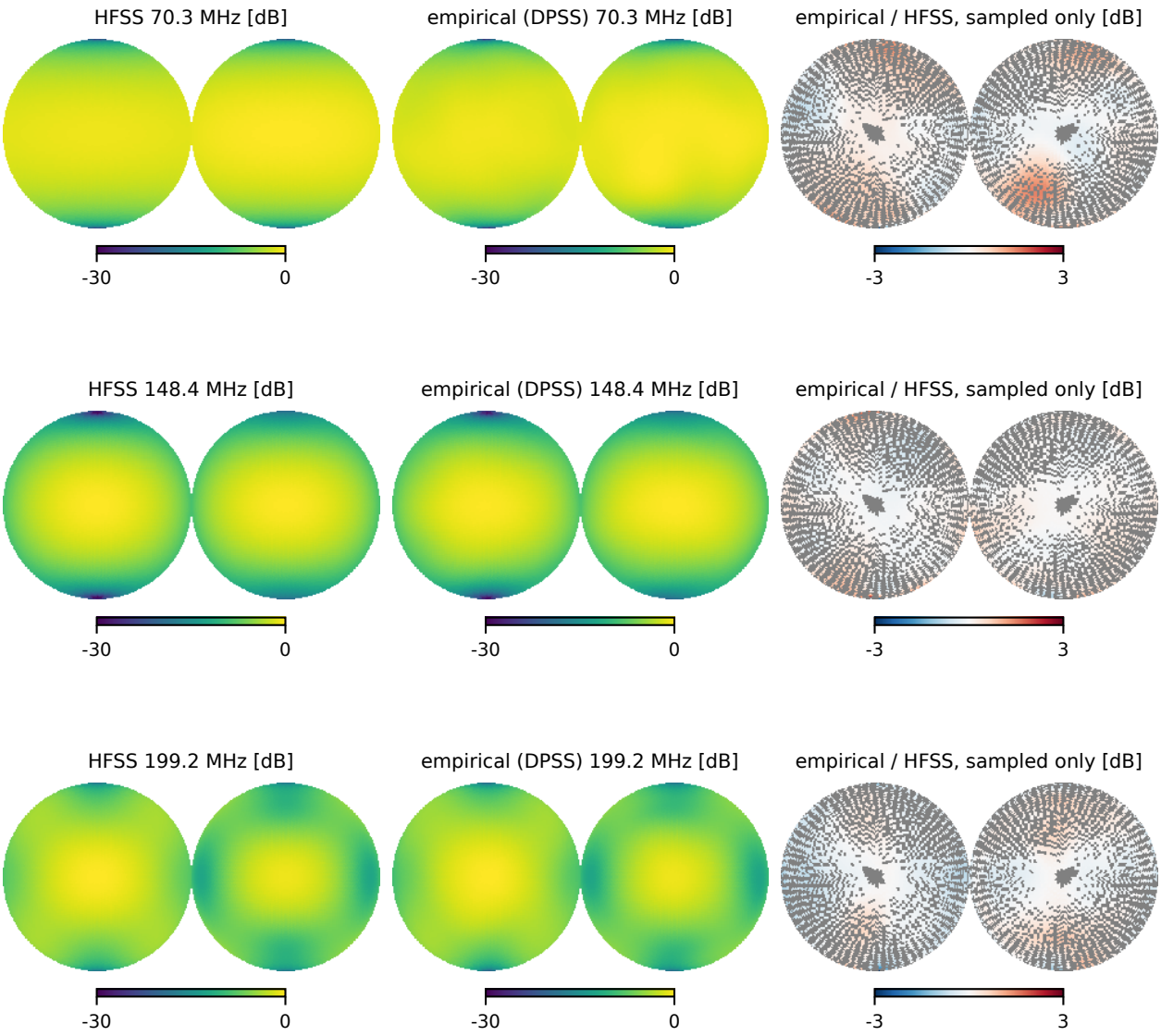


Figure 10: On the sampled pixels the exported beam differs from HFSS by 0.1–0.3 dB rms (weighted by visits and power). Total power $|\mathbf{E}|^2$ of HFSS (left) and the exported DPSS beam (centre), each peak-normalized, and their ratio on the sampled pixels only (right), at three frequencies. Each panel shows the forward hemisphere (boresight at the centre of the left disc) and the backward hemisphere (right disc).

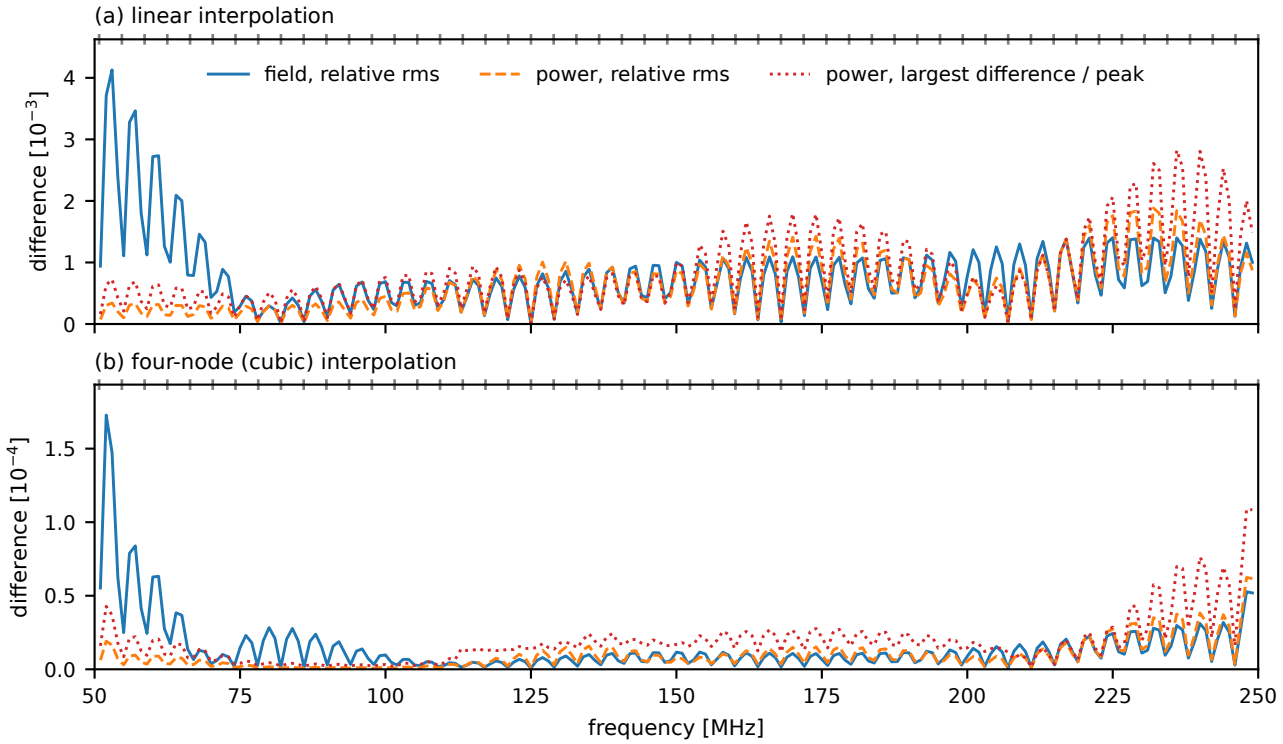


Figure 11: Interpolating the 52-frequency HFSS beam in frequency reproduces the 1 MHz solve to at most 0.28% of the peak power (linear) and 1.1×10^{-4} (four-node), far below the beam fit's held-out residual. At each integer frequency from 51 to 249 MHz, the 52-frequency fields interpolated linearly (a) and with four-node Lagrange weights (b) are compared with the 1 MHz solve of the same model, both scaled to unit mean power and made symmetric about the x mirror plane: the field's relative rms over the sphere (solid), the power's (dashed), and the largest power difference in any direction over the peak power (dotted). Note the different vertical scales. Ticks along the top mark the 52 solve frequencies.

B Raster maps of every candidate tooth

All 101 candidate teeth in frequency order, 26 pages of four, with the panels, geometry and colour scales of Fig. 7. Each row's label says whether the tooth was fitted or why it was excluded, and gives its HFSS test-stripe fractional rms. Teeth in the FM band (87–108.5 MHz) carry broadcast signals as well as the transmitter, and the background there is fitted without the FM channels. Where the fitted gain is zero (gains are constrained non-negative) there is no HFSS peak to divide by; those rows are divided by the data's 99.5th percentile instead (marked "gain 0"), their HFSS panel sits at the floor, and the test rms is 1. Where the gain is merely small, the scaled data run far off the top of the scale.

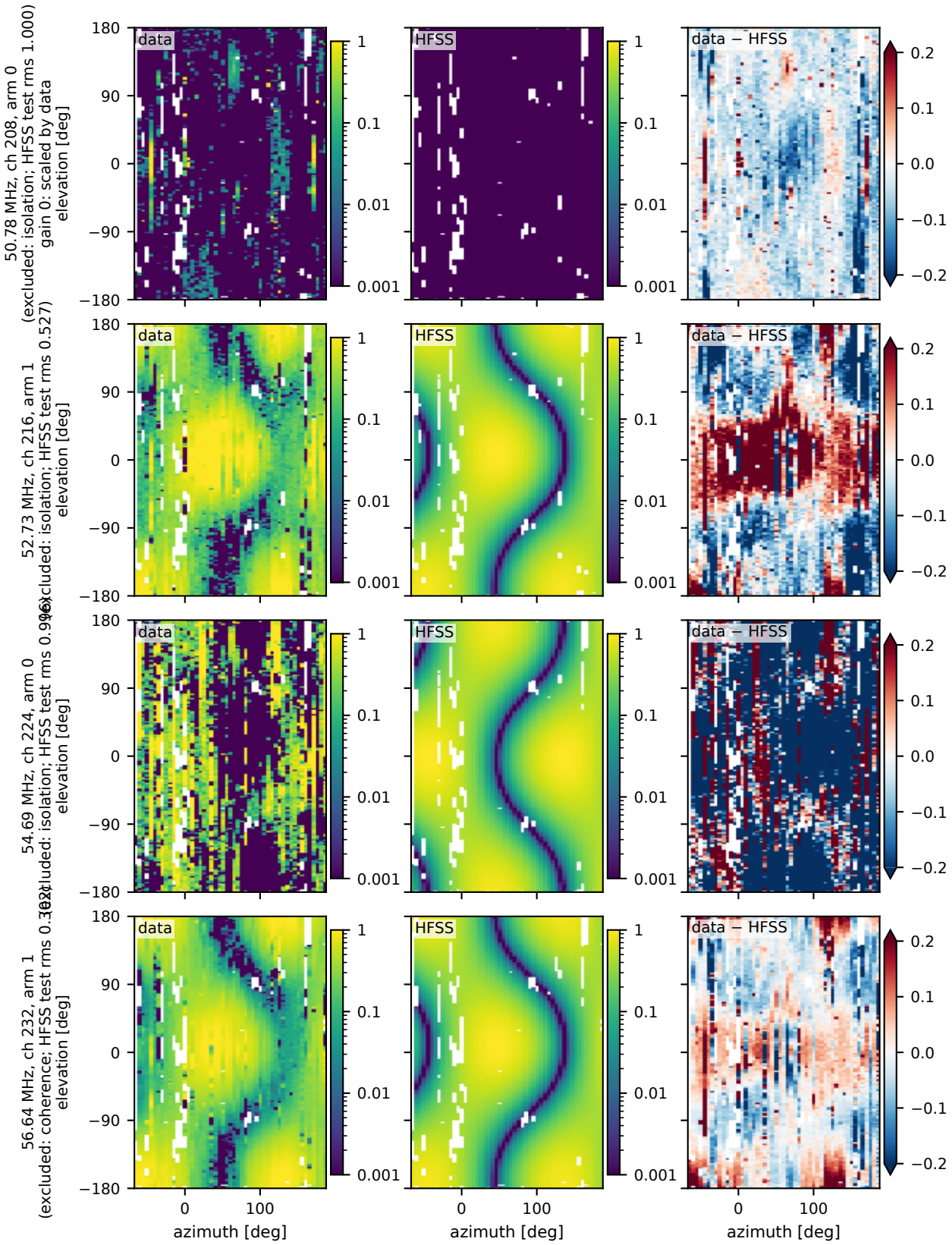


Figure 12: Raster maps, 50.78–56.64 MHz (page 1 of 26); panels as in Fig. 7.

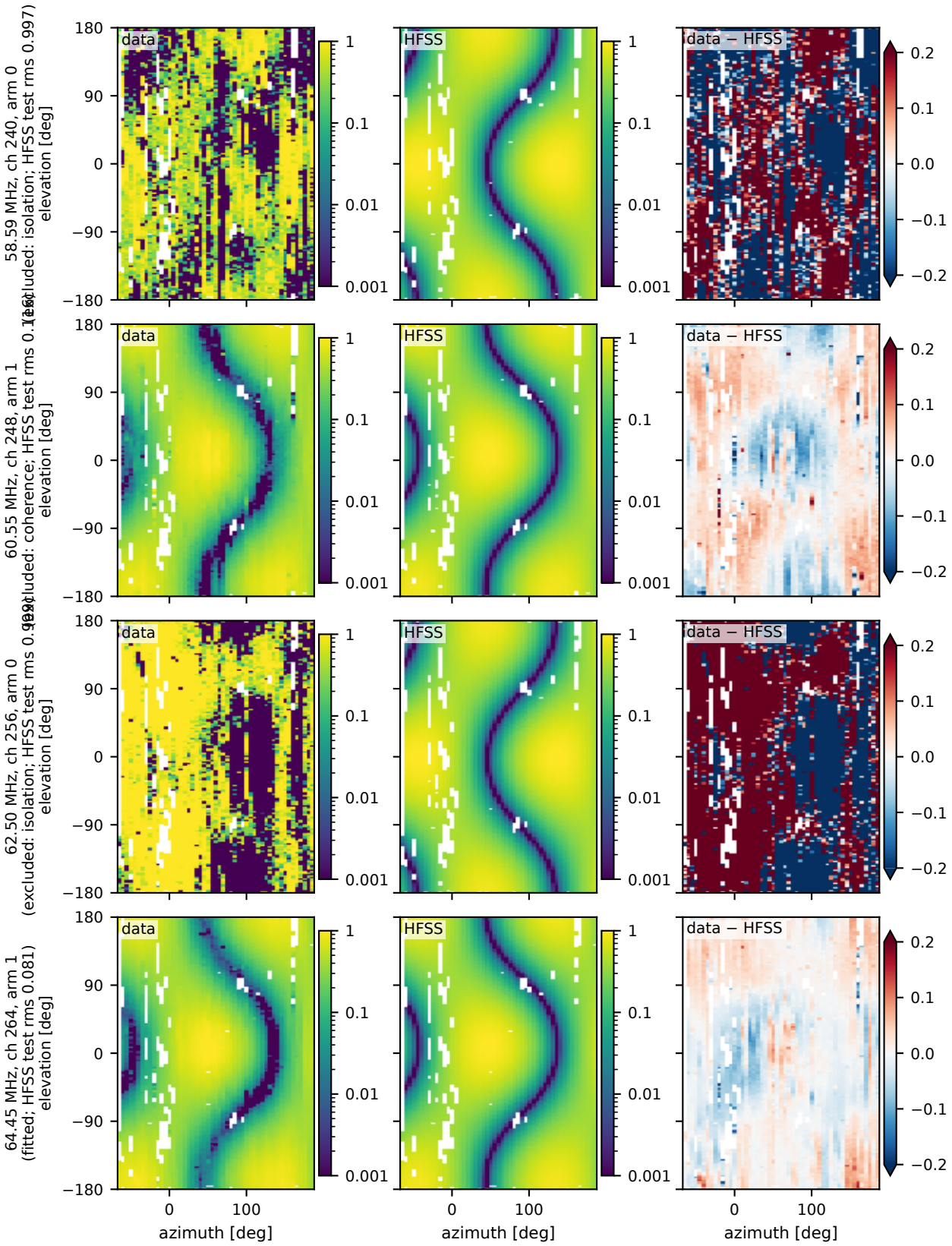


Figure 13: Raster maps, 58.59–64.45 MHz (page 2 of 26); panels as in Fig. 7.

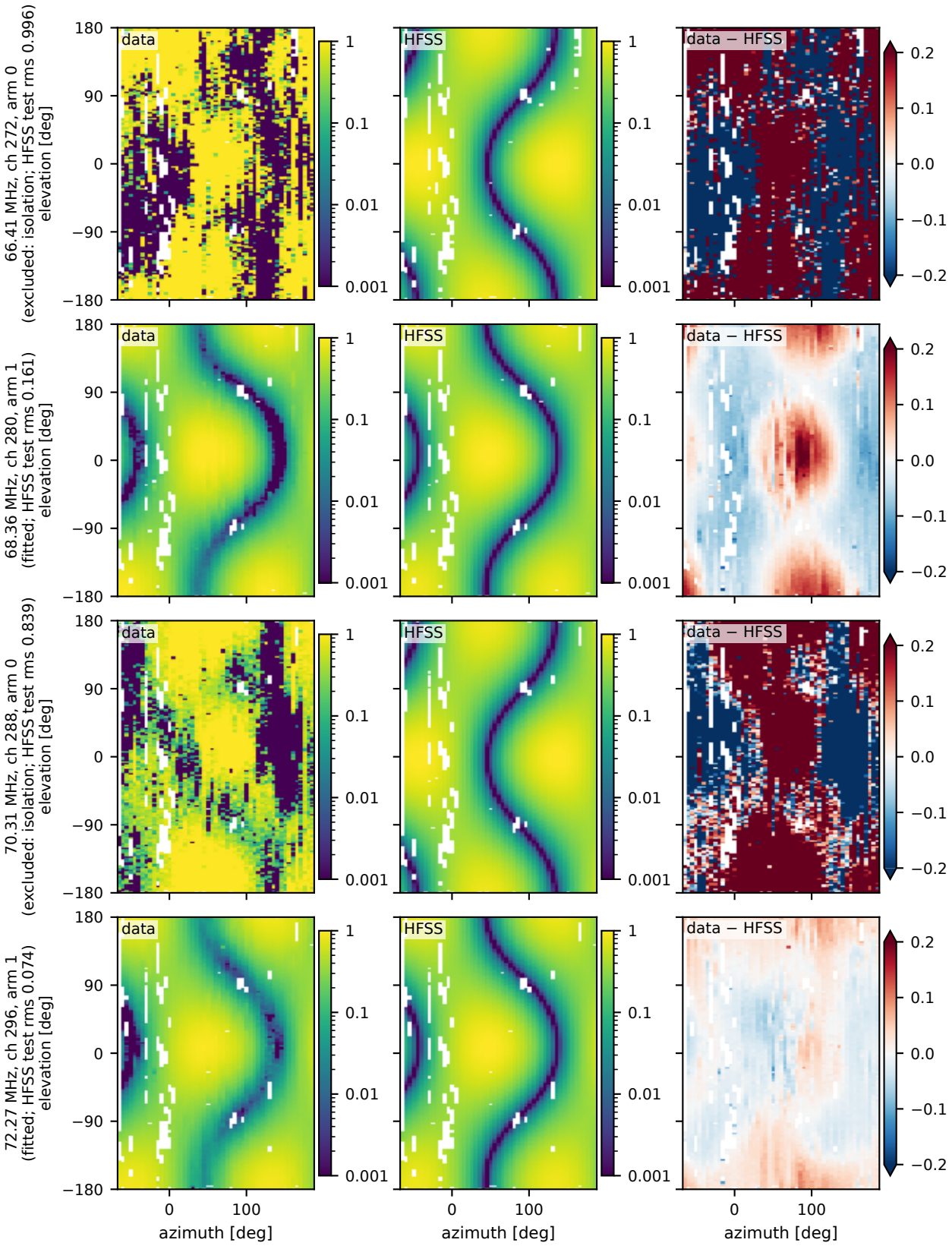


Figure 14: Raster maps, 66.41–72.27 MHz (page 3 of 26); panels as in Fig. 7.

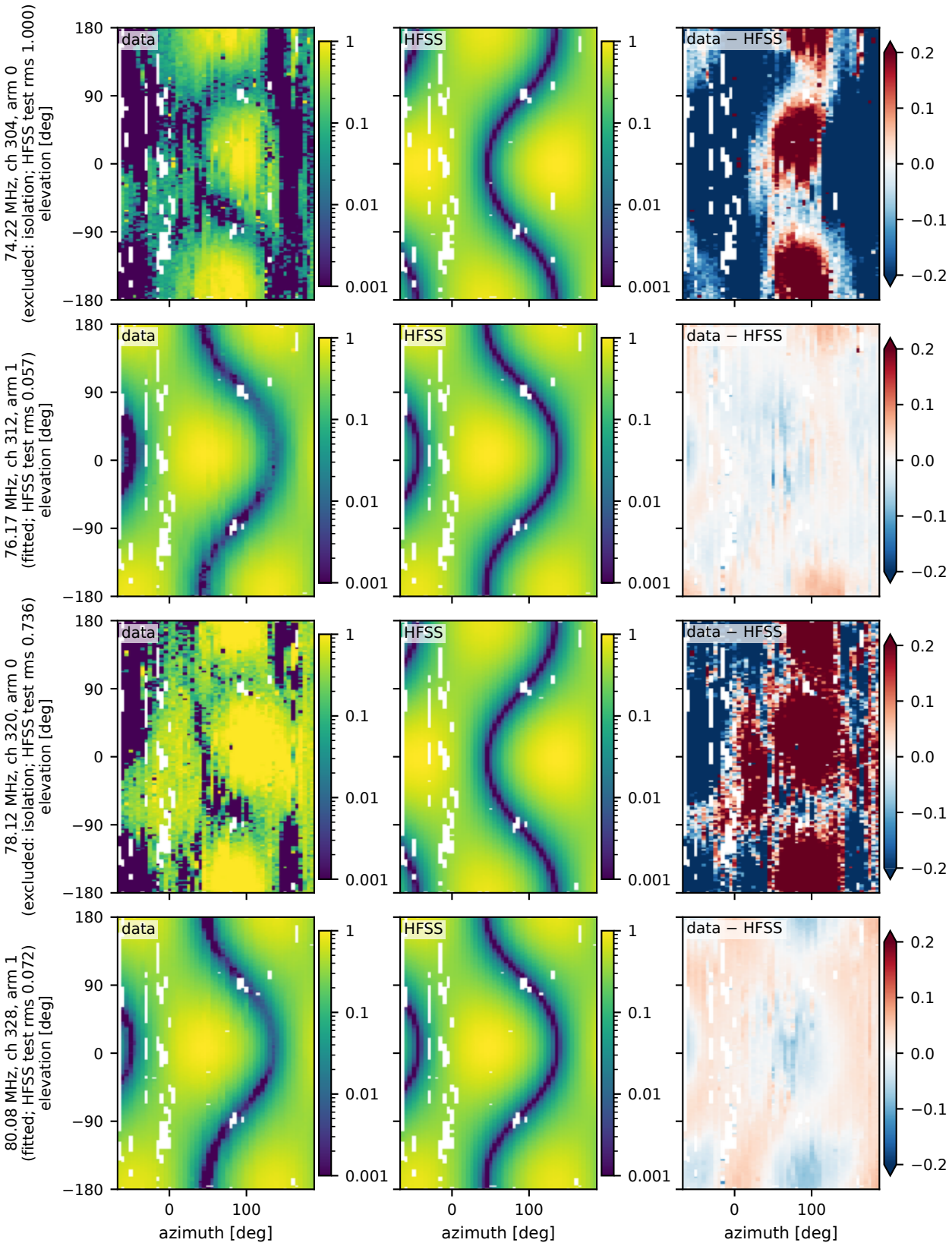


Figure 15: Raster maps, 74.22–80.08 MHz (page 4 of 26); panels as in Fig. 7.

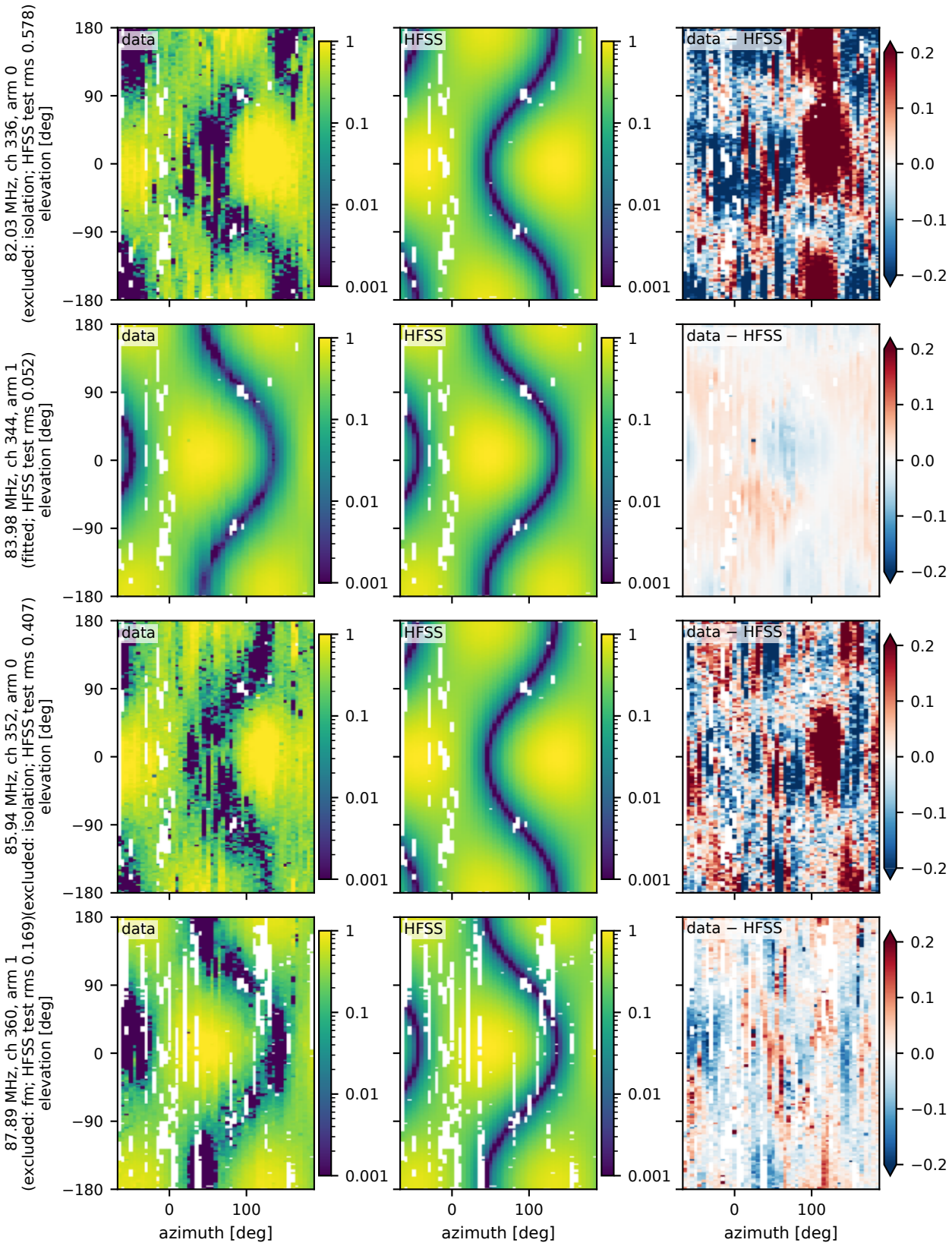


Figure 16: Raster maps, 82.03–87.89 MHz (page 5 of 26); panels as in Fig. 7.

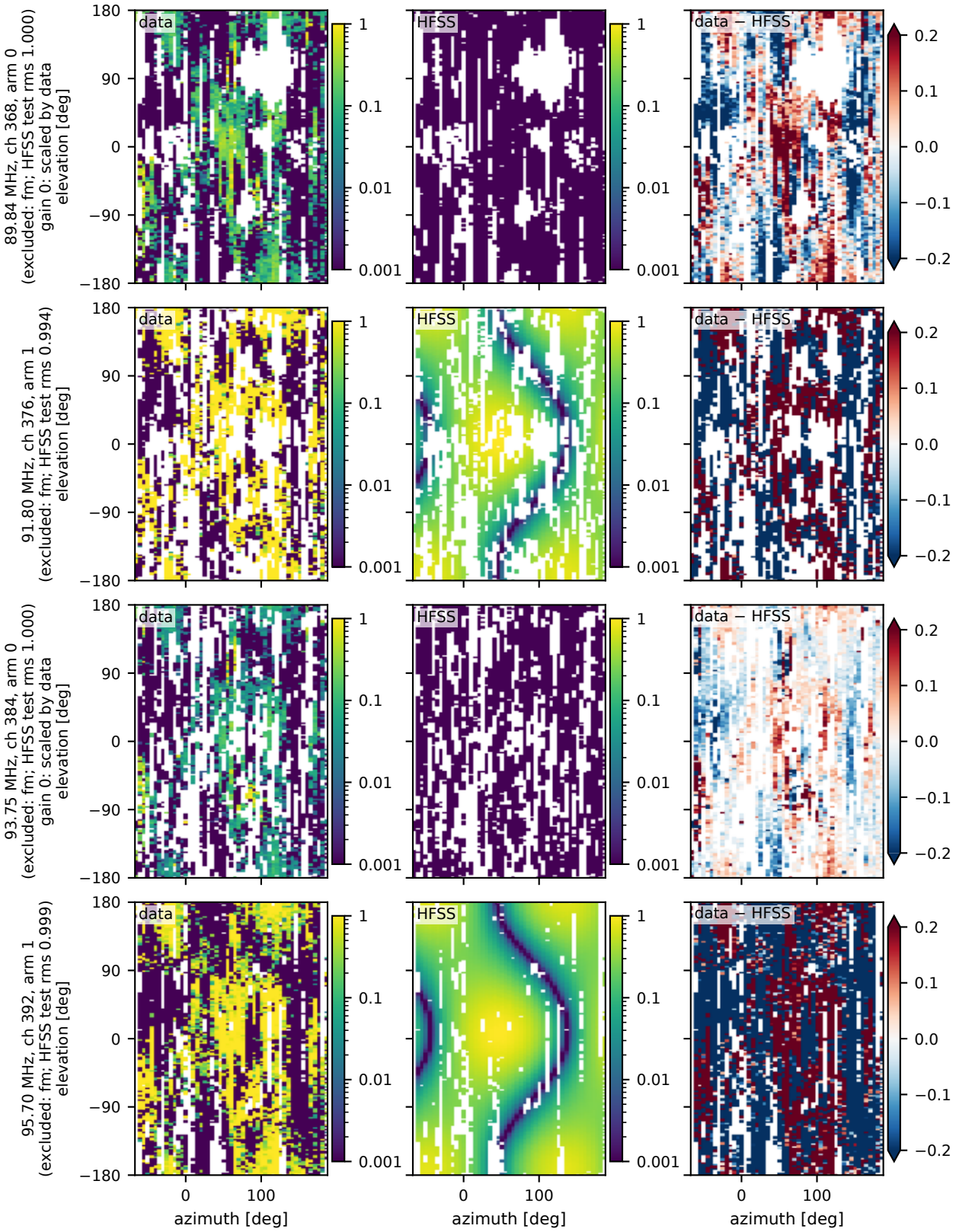


Figure 17: Raster maps, 89.84–95.70 MHz (page 6 of 26); panels as in Fig. 7.

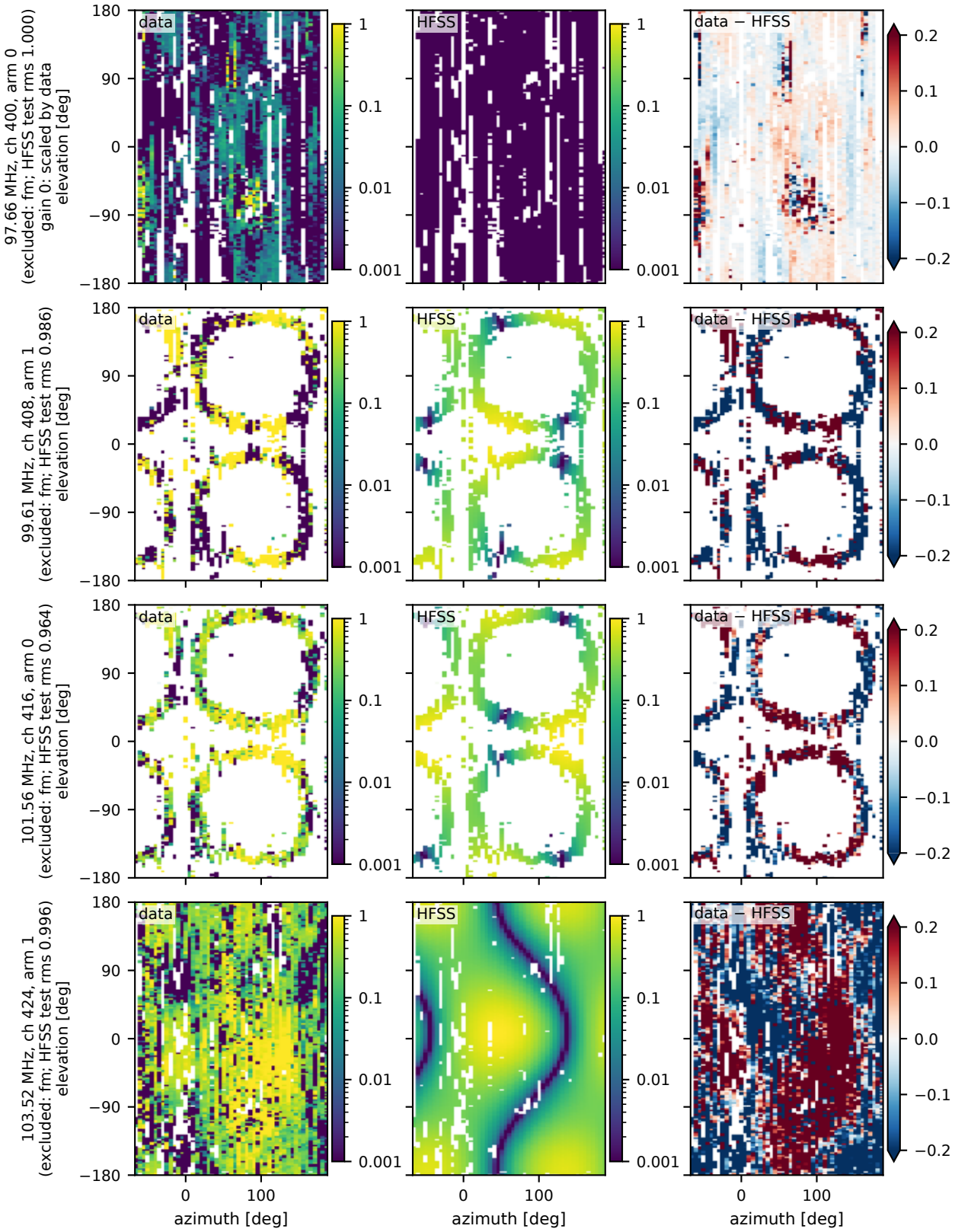


Figure 18: Raster maps, 97.66–103.52 MHz (page 7 of 26); panels as in Fig. 7.

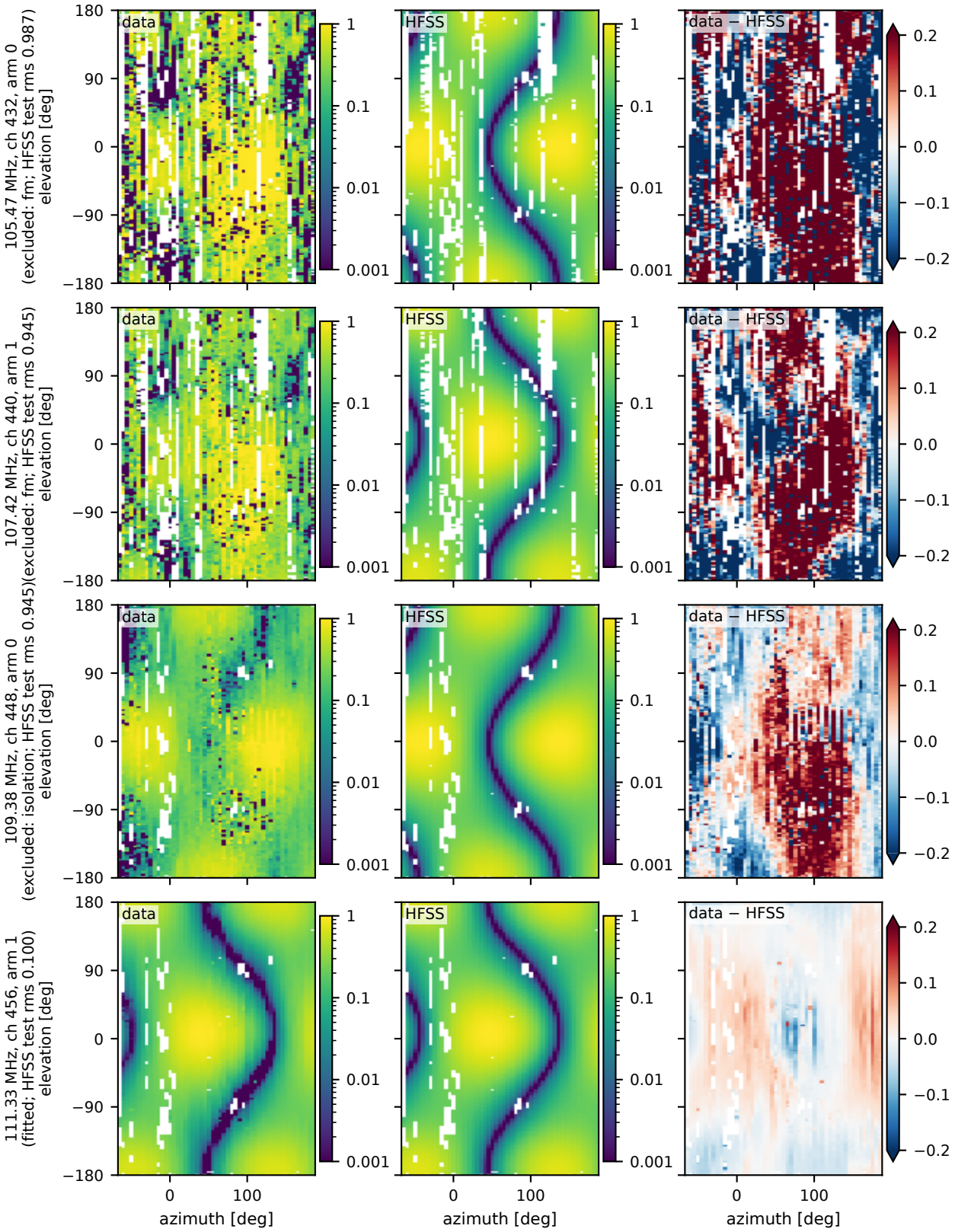


Figure 19: Raster maps, 105.47–111.33 MHz (page 8 of 26); panels as in Fig. 7.

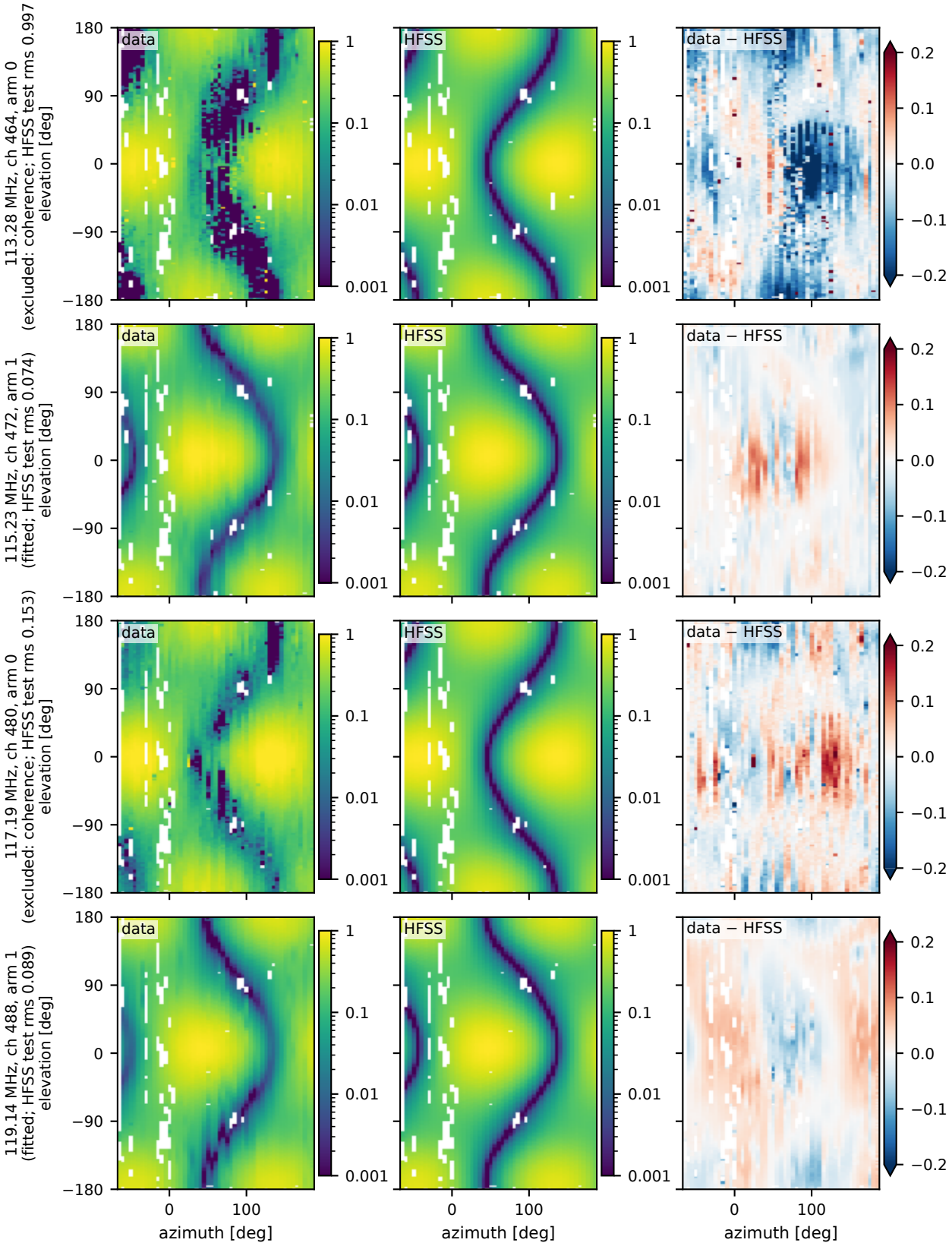


Figure 20: Raster maps, 113.28–119.14 MHz (page 9 of 26); panels as in Fig. 7.

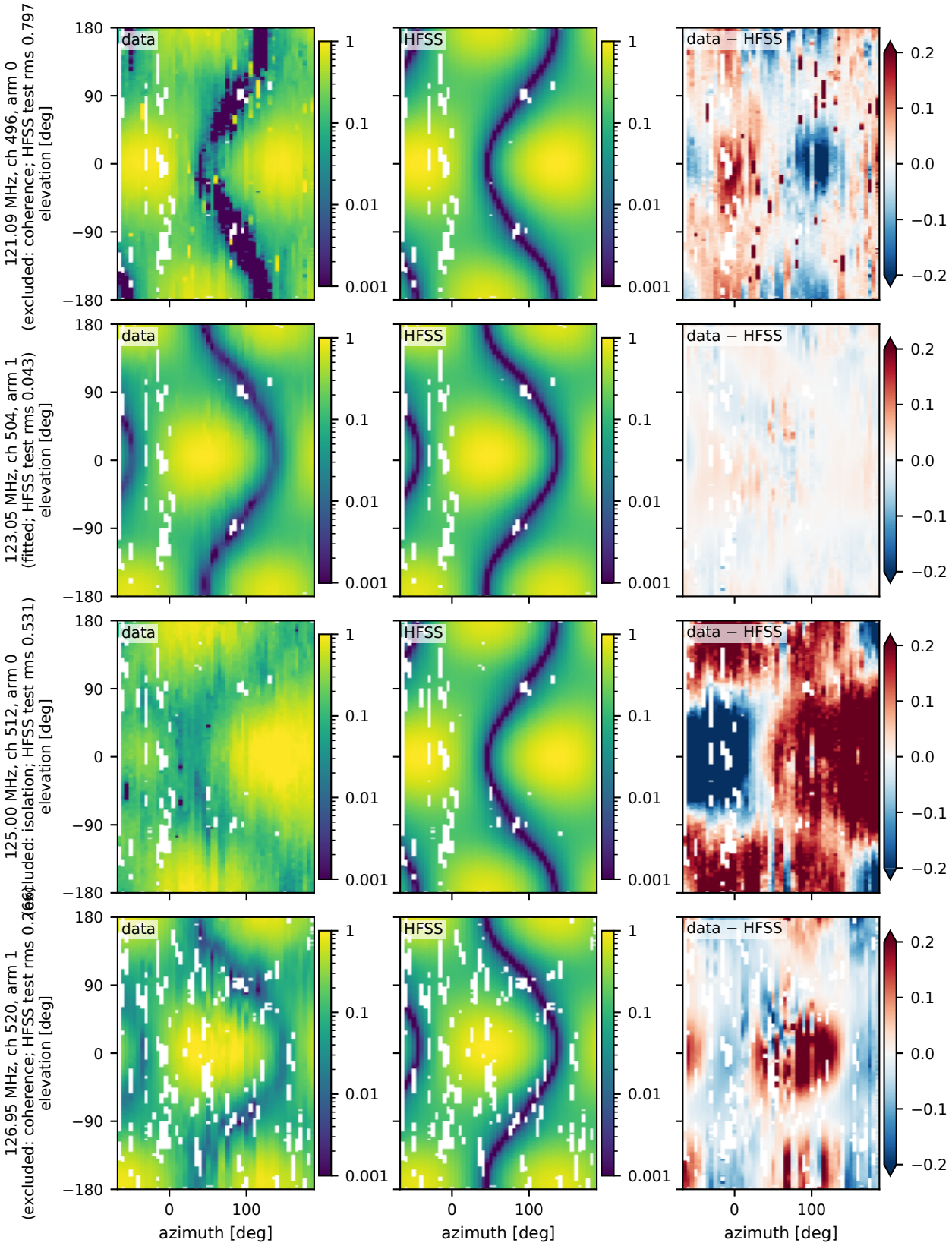


Figure 21: Raster maps, 121.09–126.95 MHz (page 10 of 26); panels as in Fig. 7.

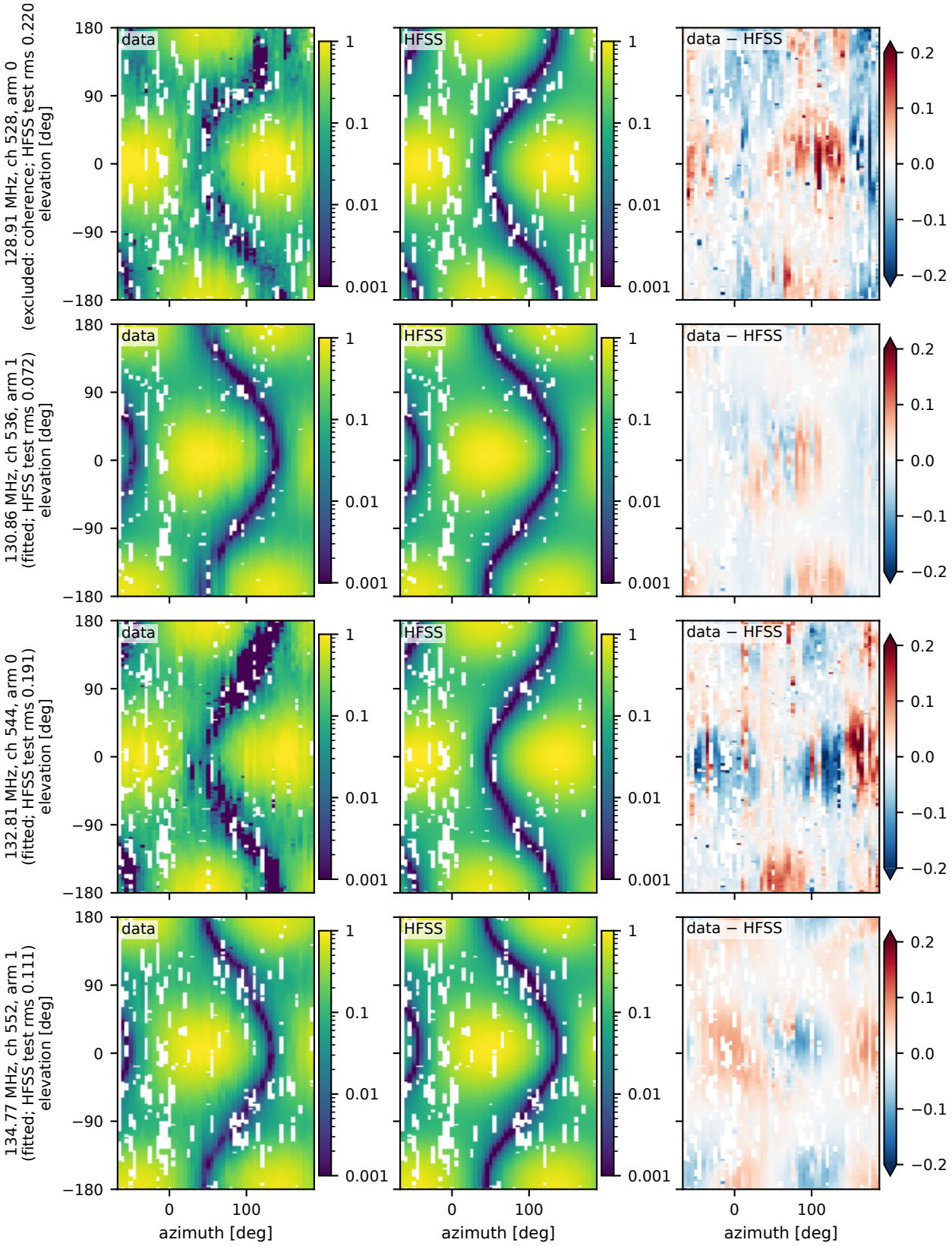


Figure 22: Raster maps, 128.91–134.77 MHz (page 11 of 26); panels as in Fig. 7.

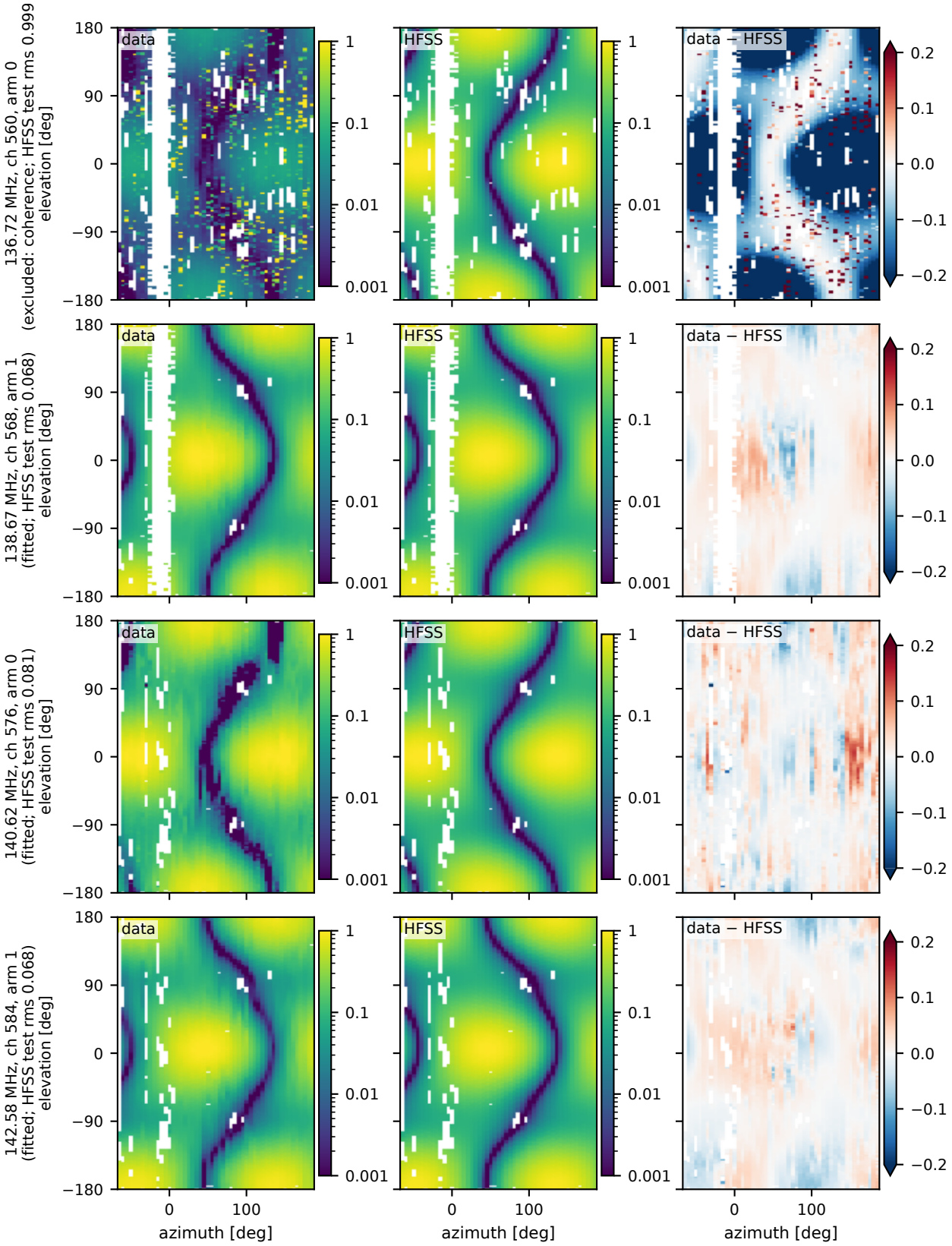


Figure 23: Raster maps, 136.72–142.58 MHz (page 12 of 26); panels as in Fig. 7.

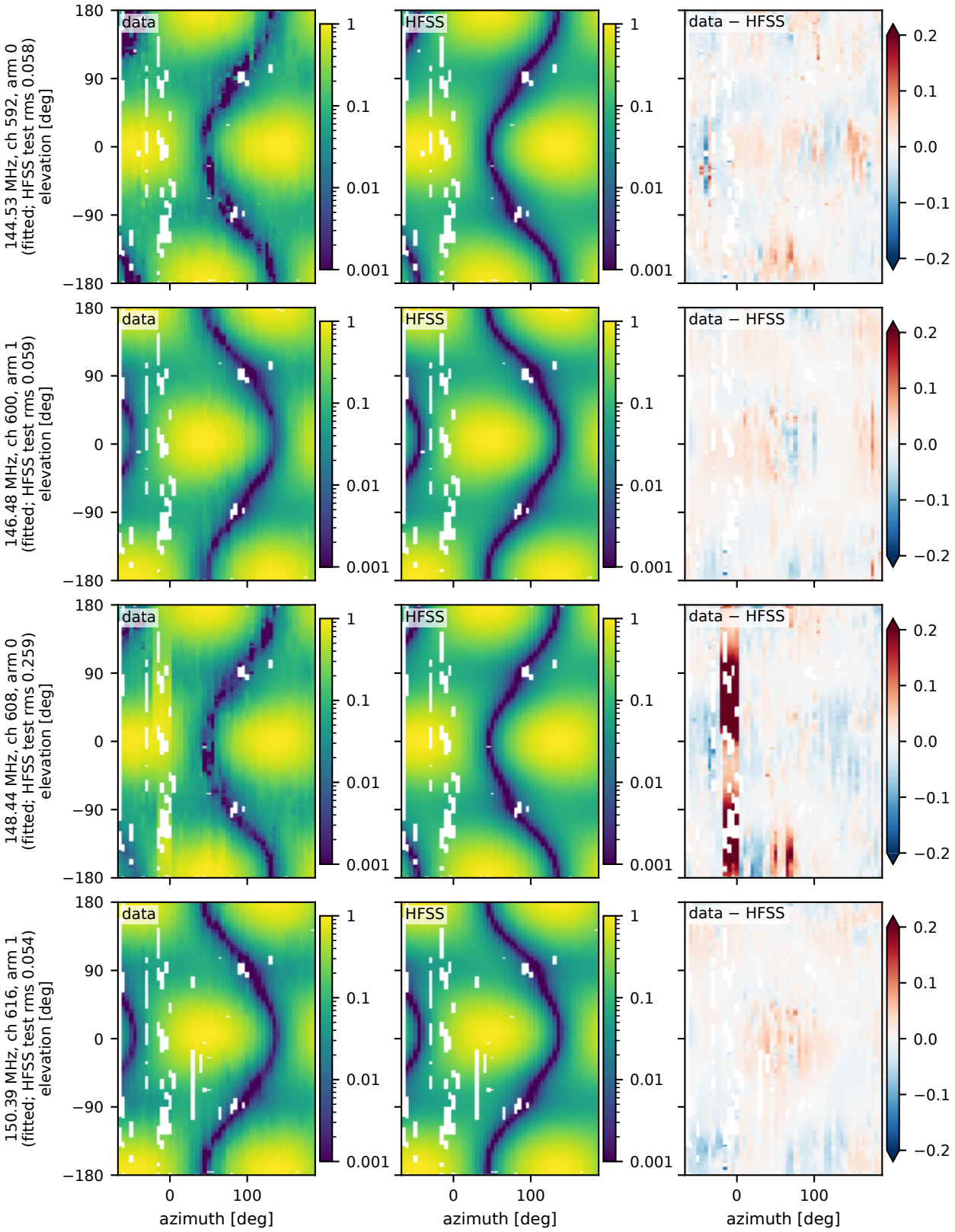


Figure 24: Raster maps, 144.53–150.39 MHz (page 13 of 26); panels as in Fig. 7.

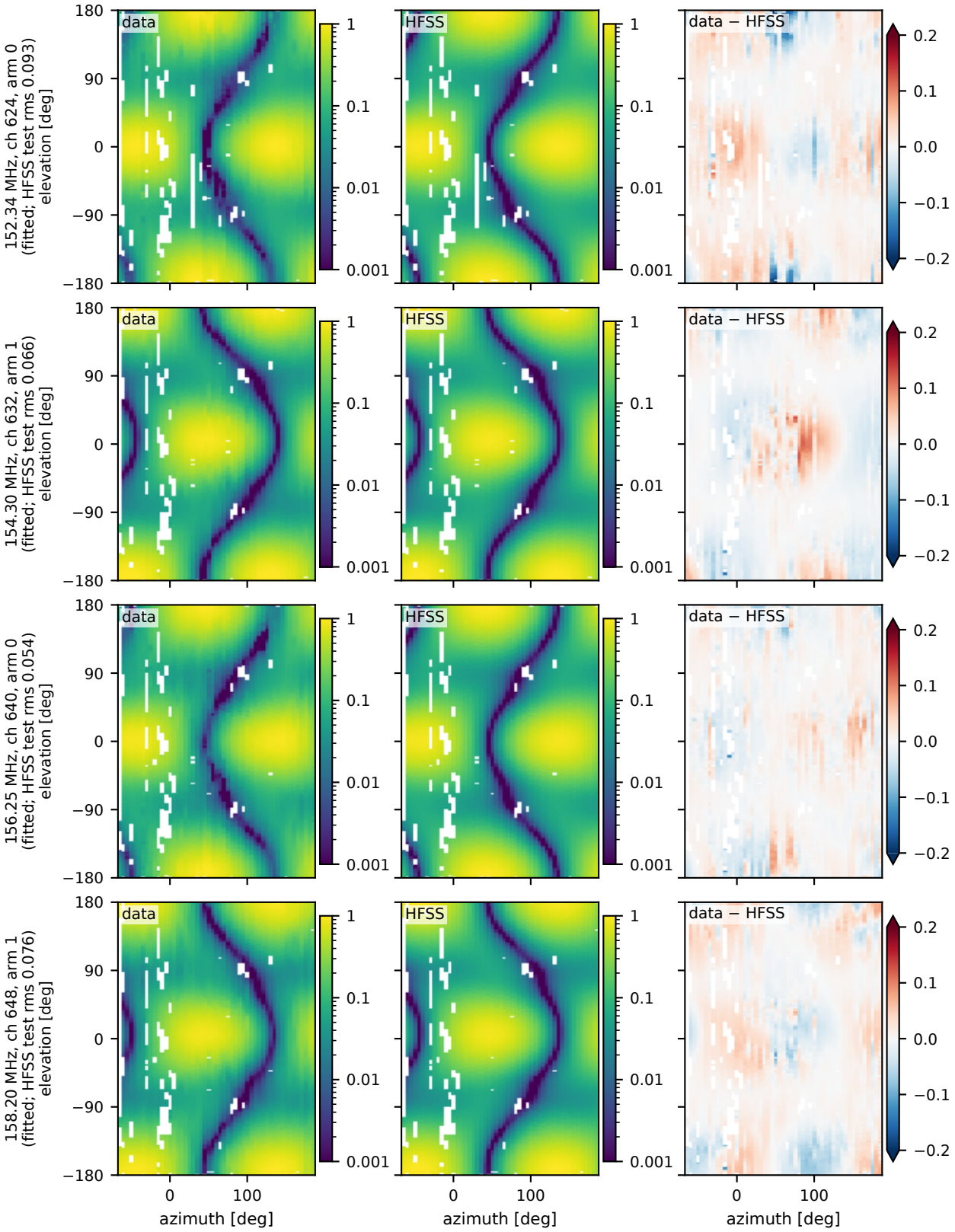


Figure 25: Raster maps, 152.34–158.20 MHz (page 14 of 26); panels as in Fig. 7.

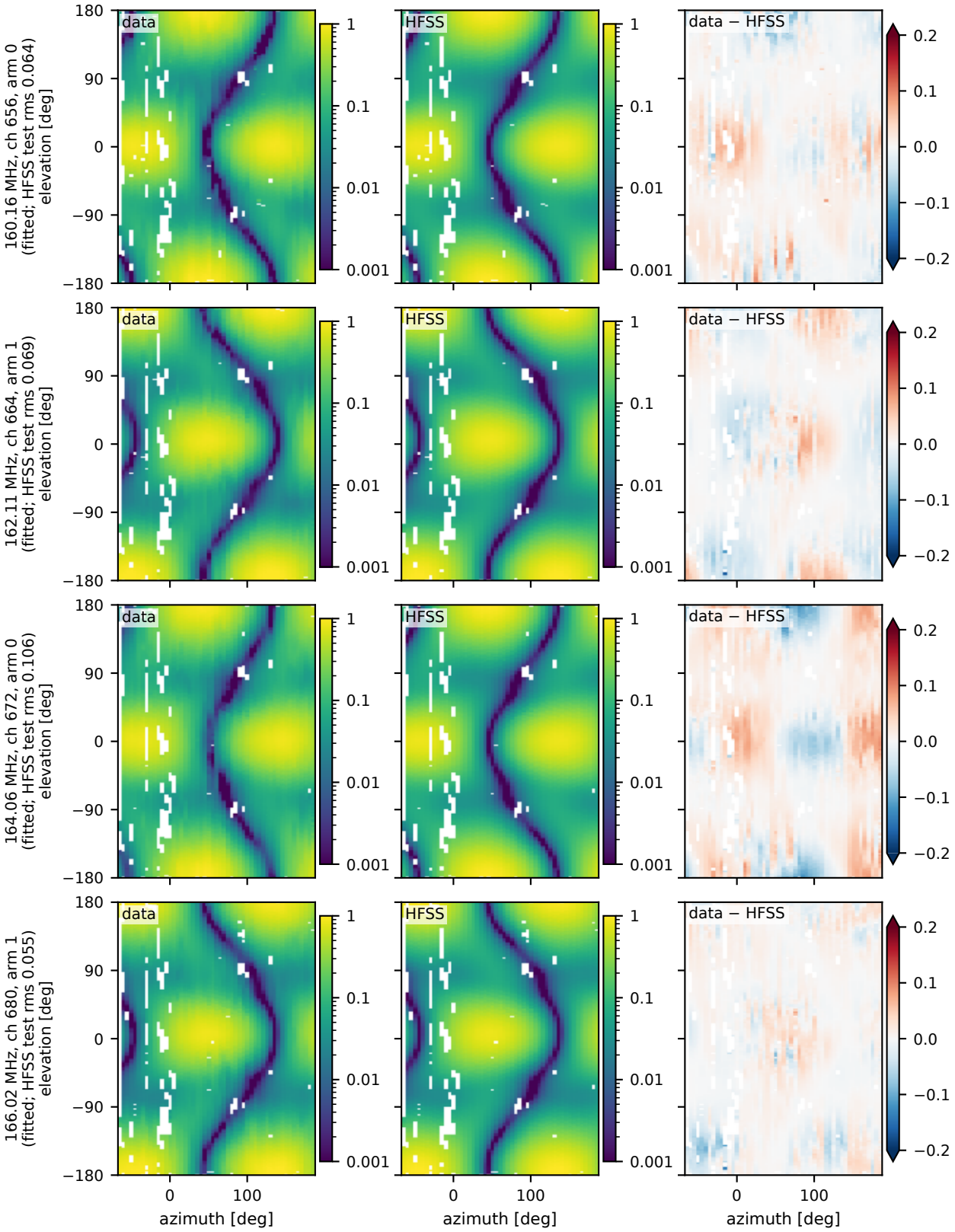


Figure 26: Raster maps, 160.16–166.02 MHz (page 15 of 26); panels as in Fig. 7.

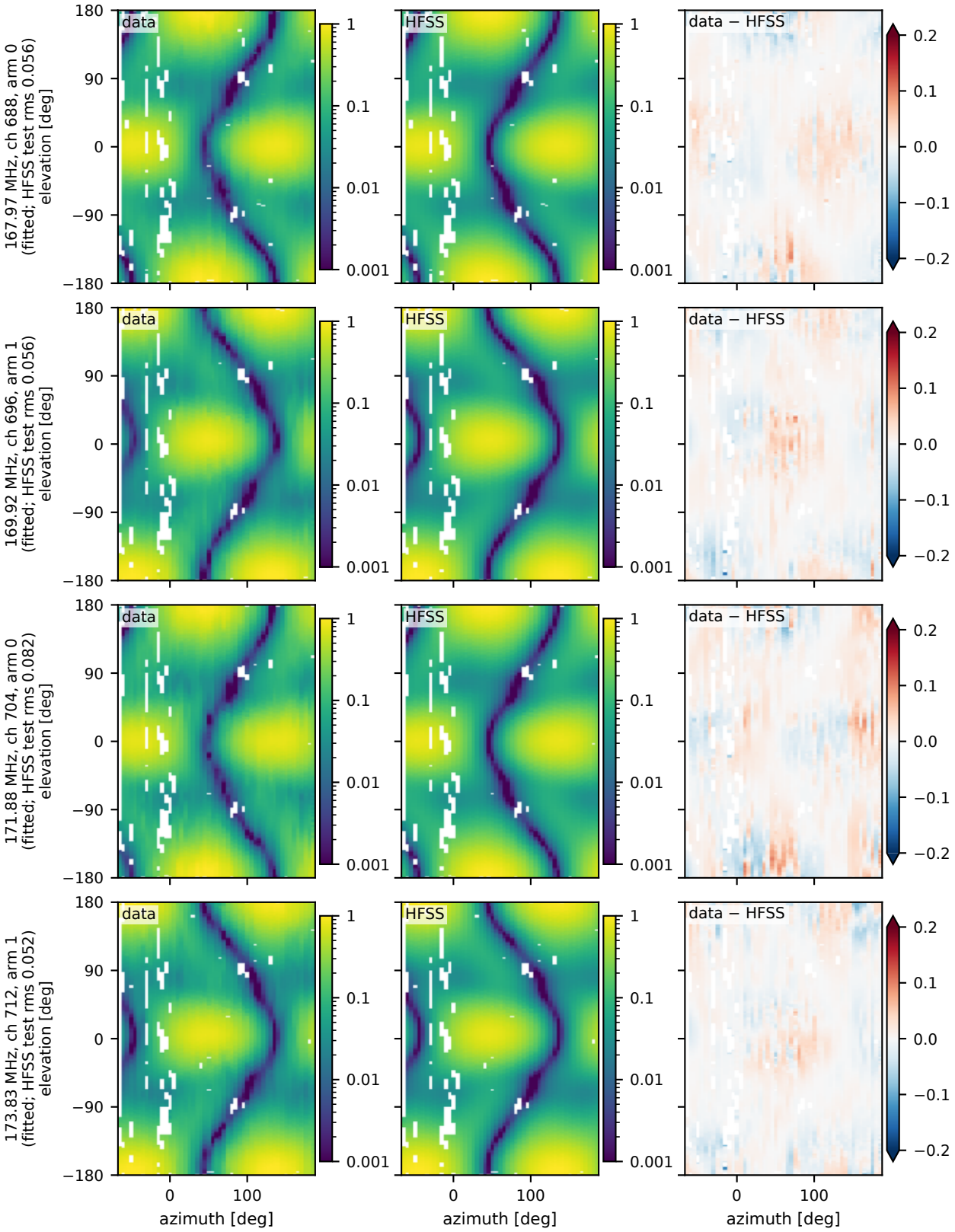


Figure 27: Raster maps, 167.97–173.83 MHz (page 16 of 26); panels as in Fig. 7.

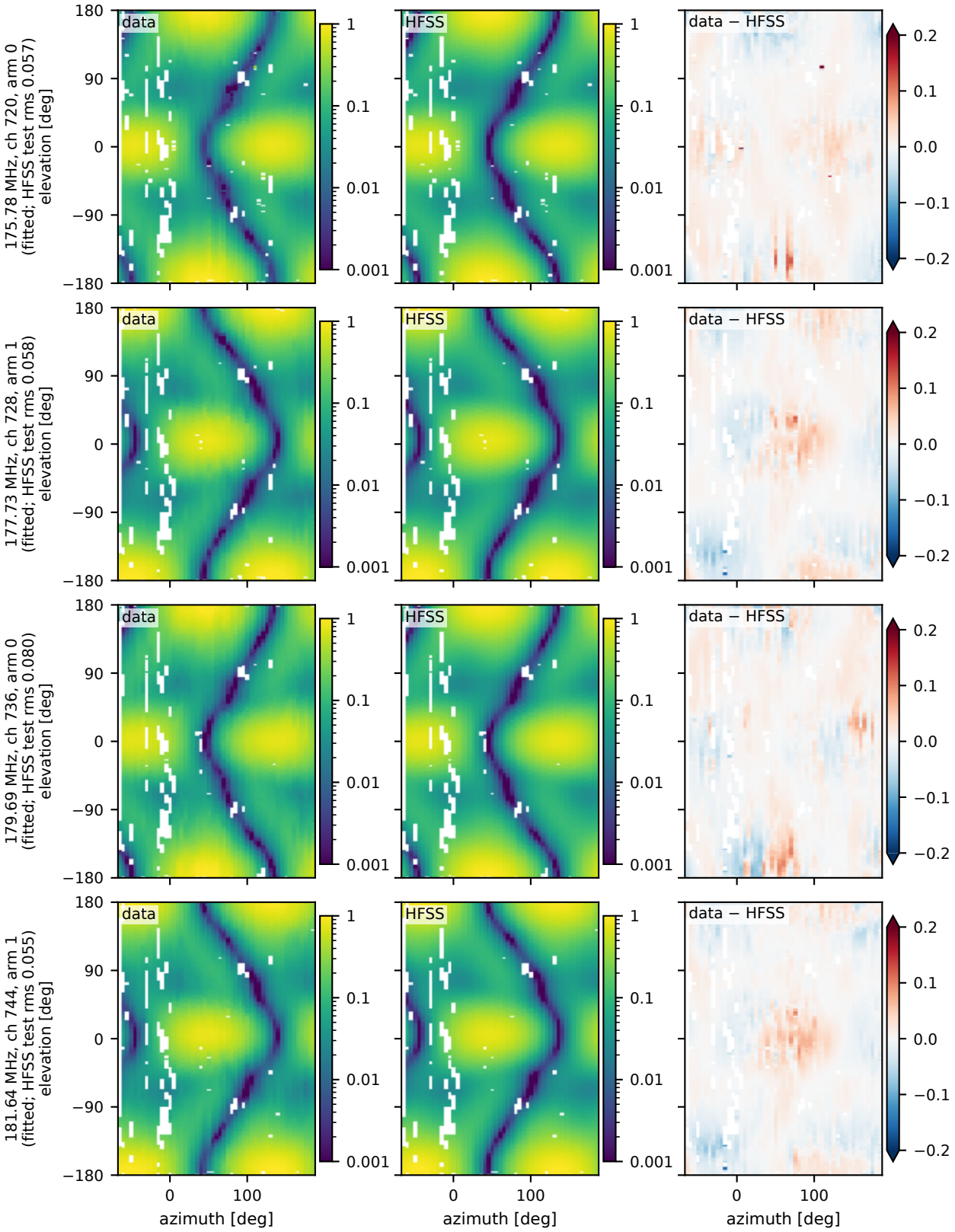


Figure 28: Raster maps, 175.78–181.64 MHz (page 17 of 26); panels as in Fig. 7.

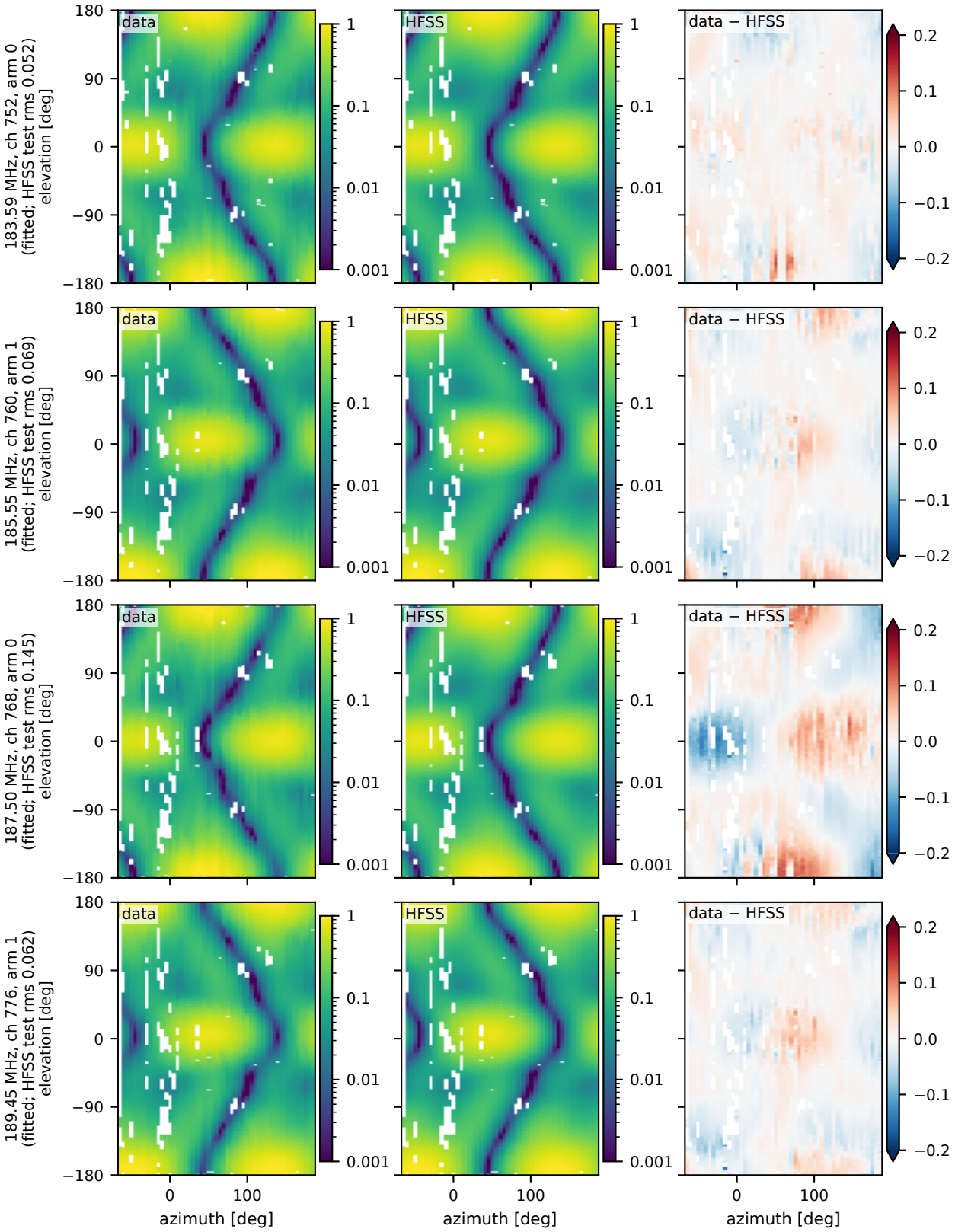


Figure 29: Raster maps, 183.59–189.45 MHz (page 18 of 26); panels as in Fig. 7.

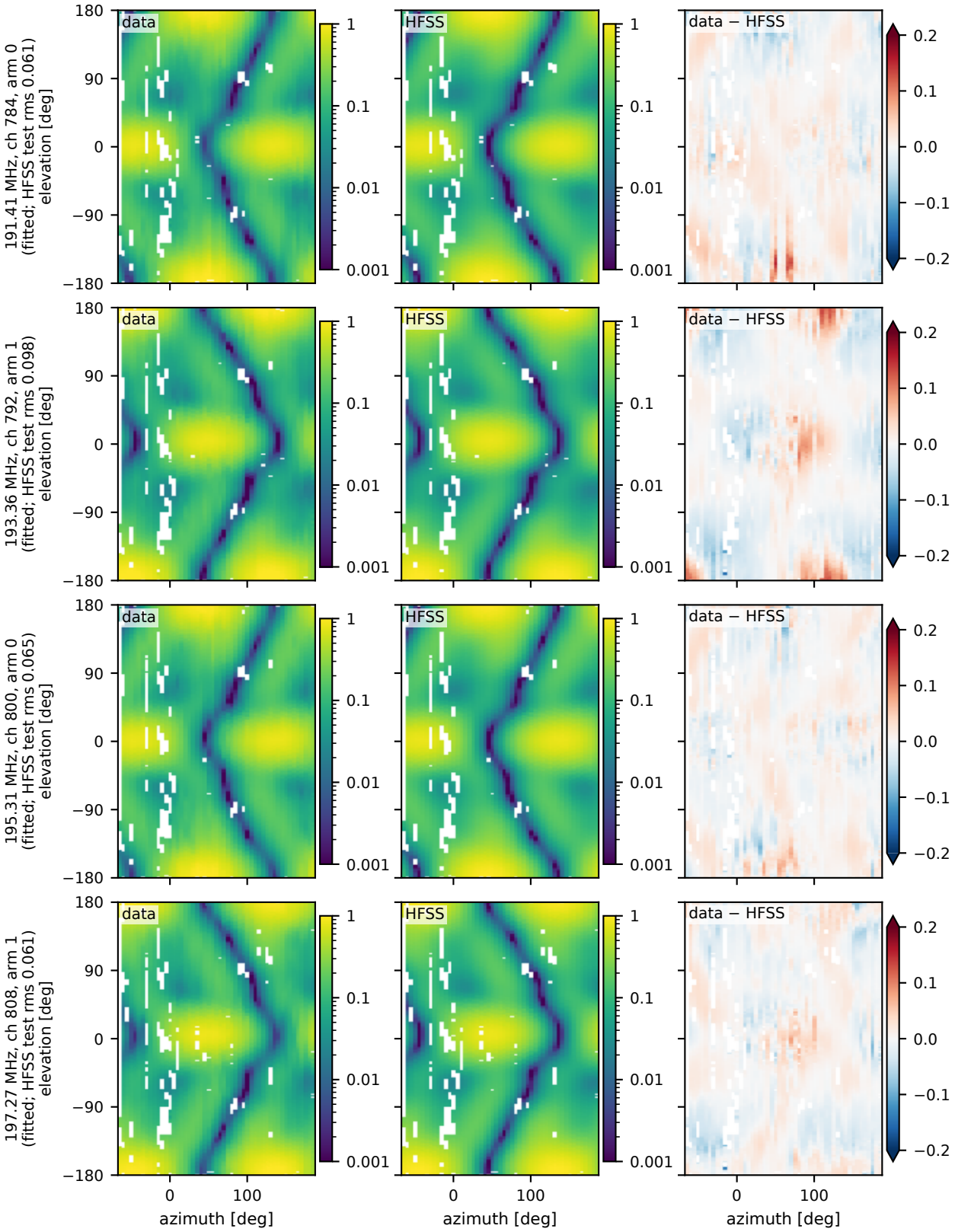


Figure 30: Raster maps, 191.41–197.27 MHz (page 19 of 26); panels as in Fig. 7.

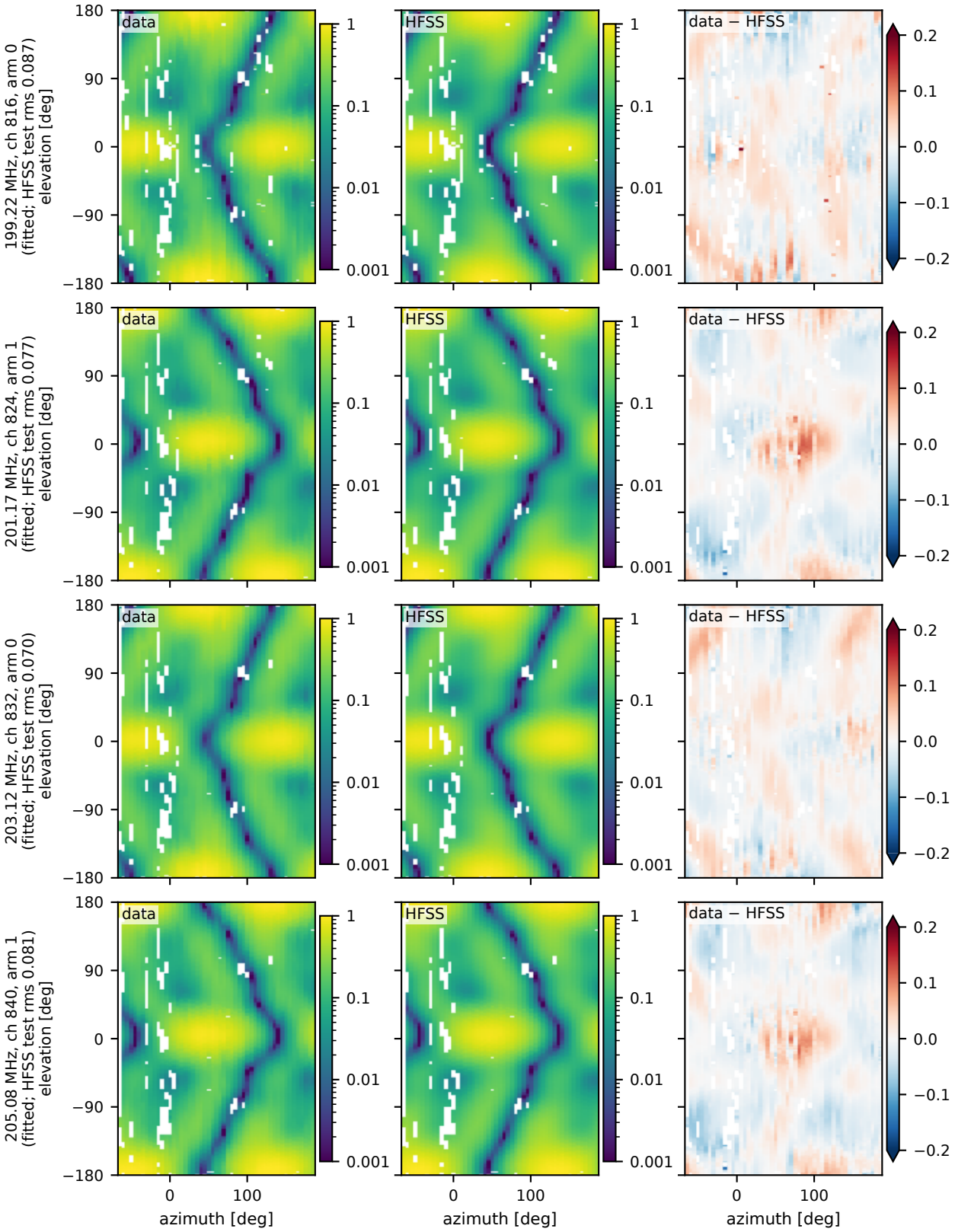


Figure 31: Raster maps, 199.22–205.08 MHz (page 20 of 26); panels as in Fig. 7.

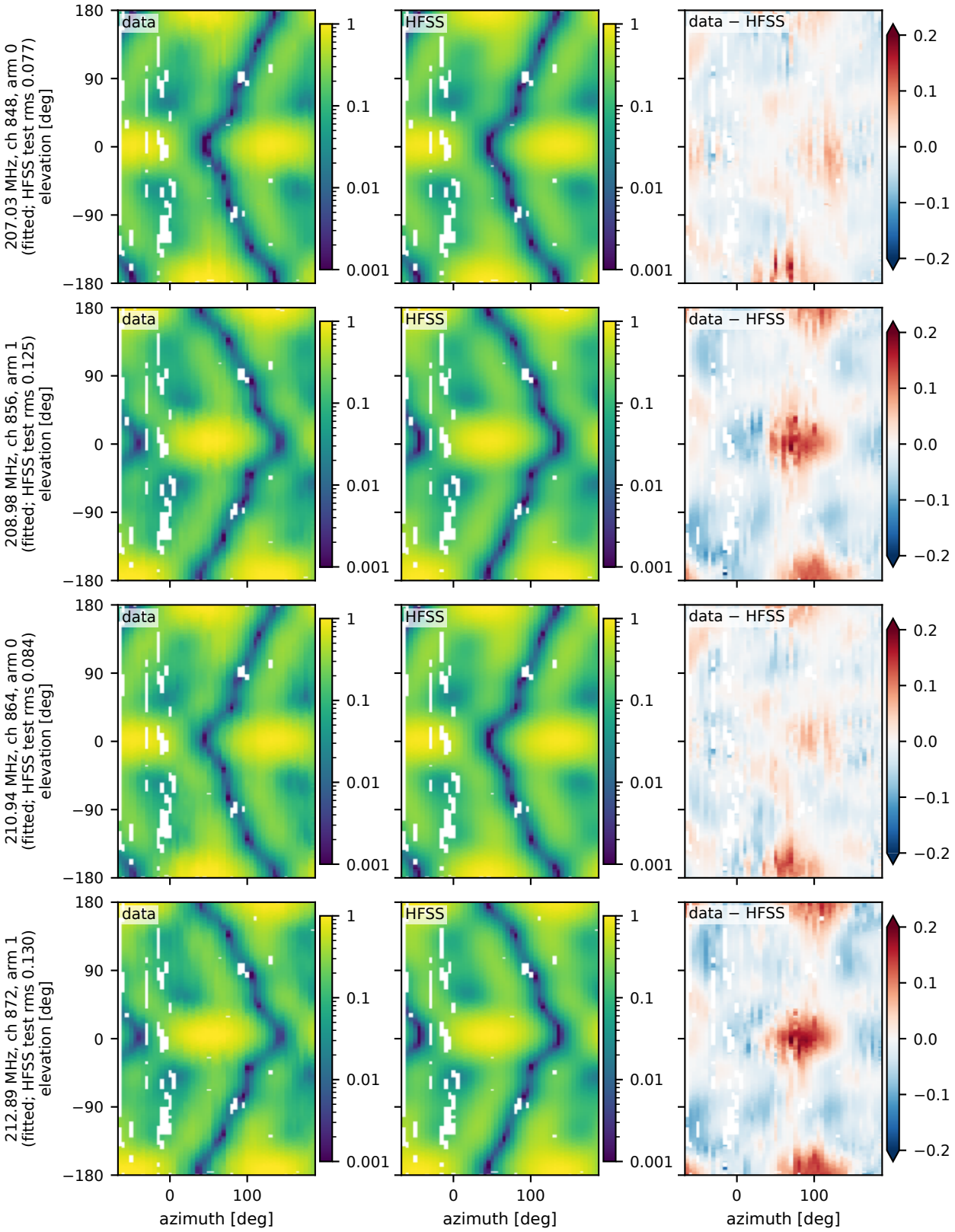


Figure 32: Raster maps, 207.03–212.89 MHz (page 21 of 26); panels as in Fig. 7.

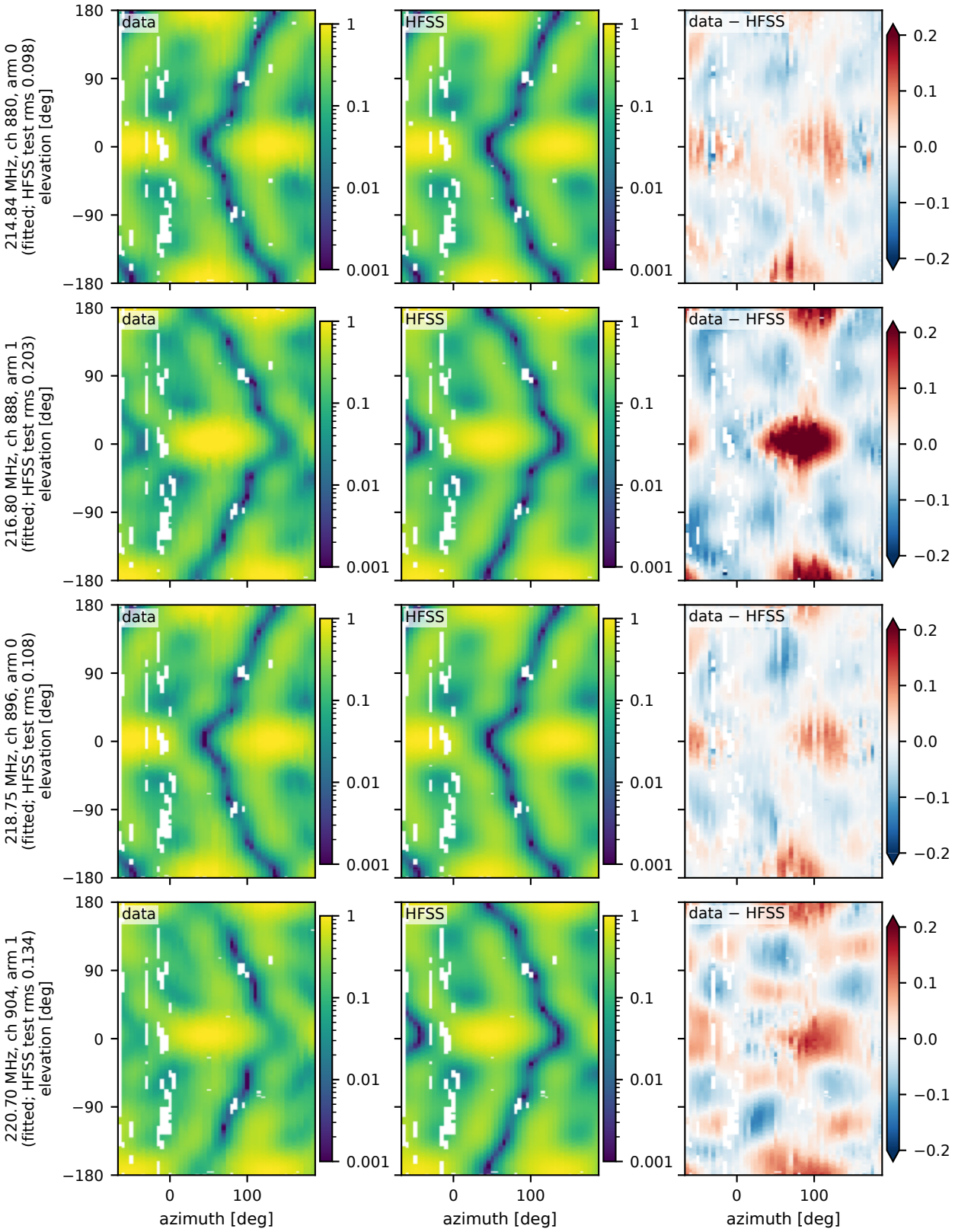


Figure 33: Raster maps, 214.84–220.70 MHz (page 22 of 26); panels as in Fig. 7.

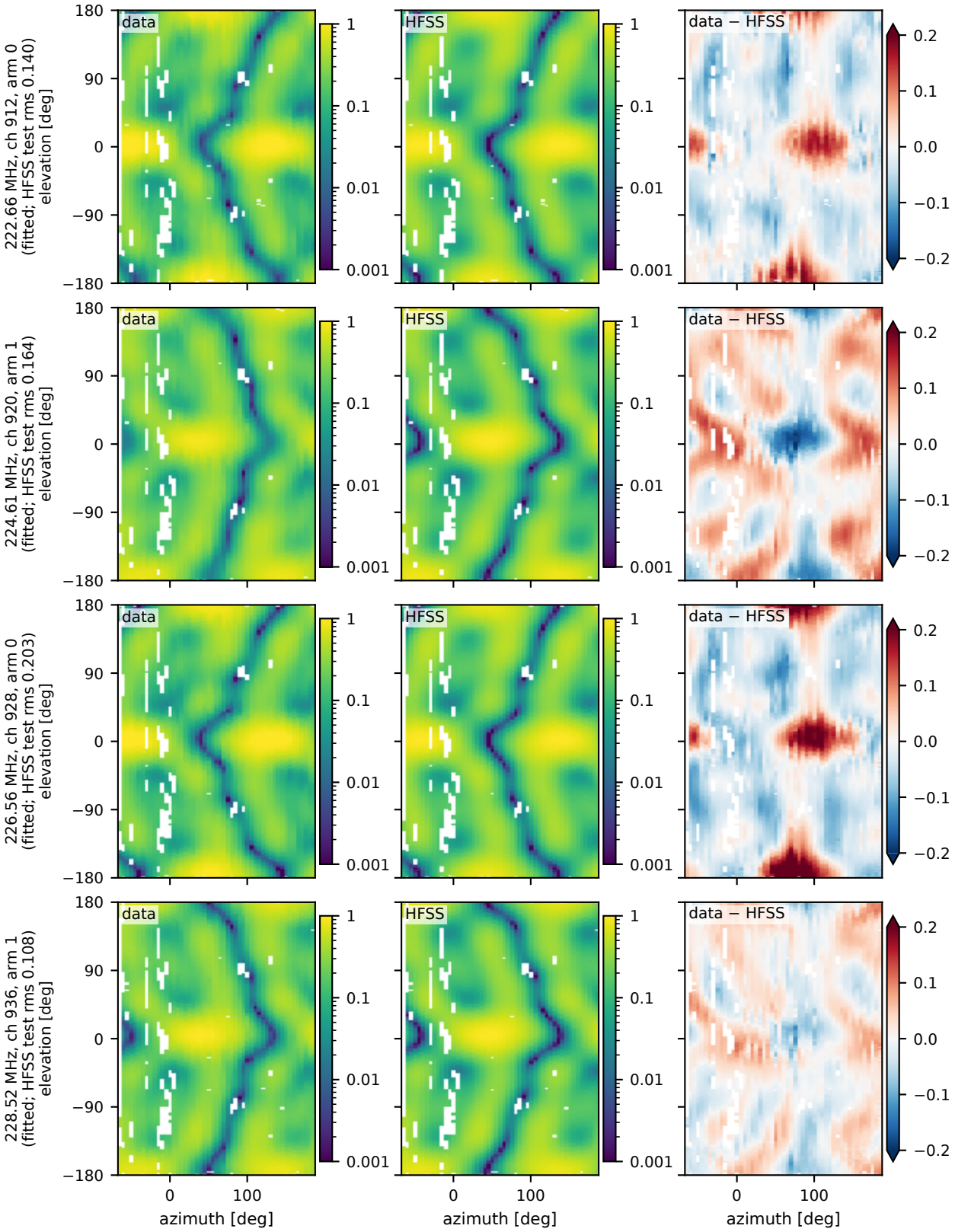


Figure 34: Raster maps, 222.66–228.52 MHz (page 23 of 26); panels as in Fig. 7.

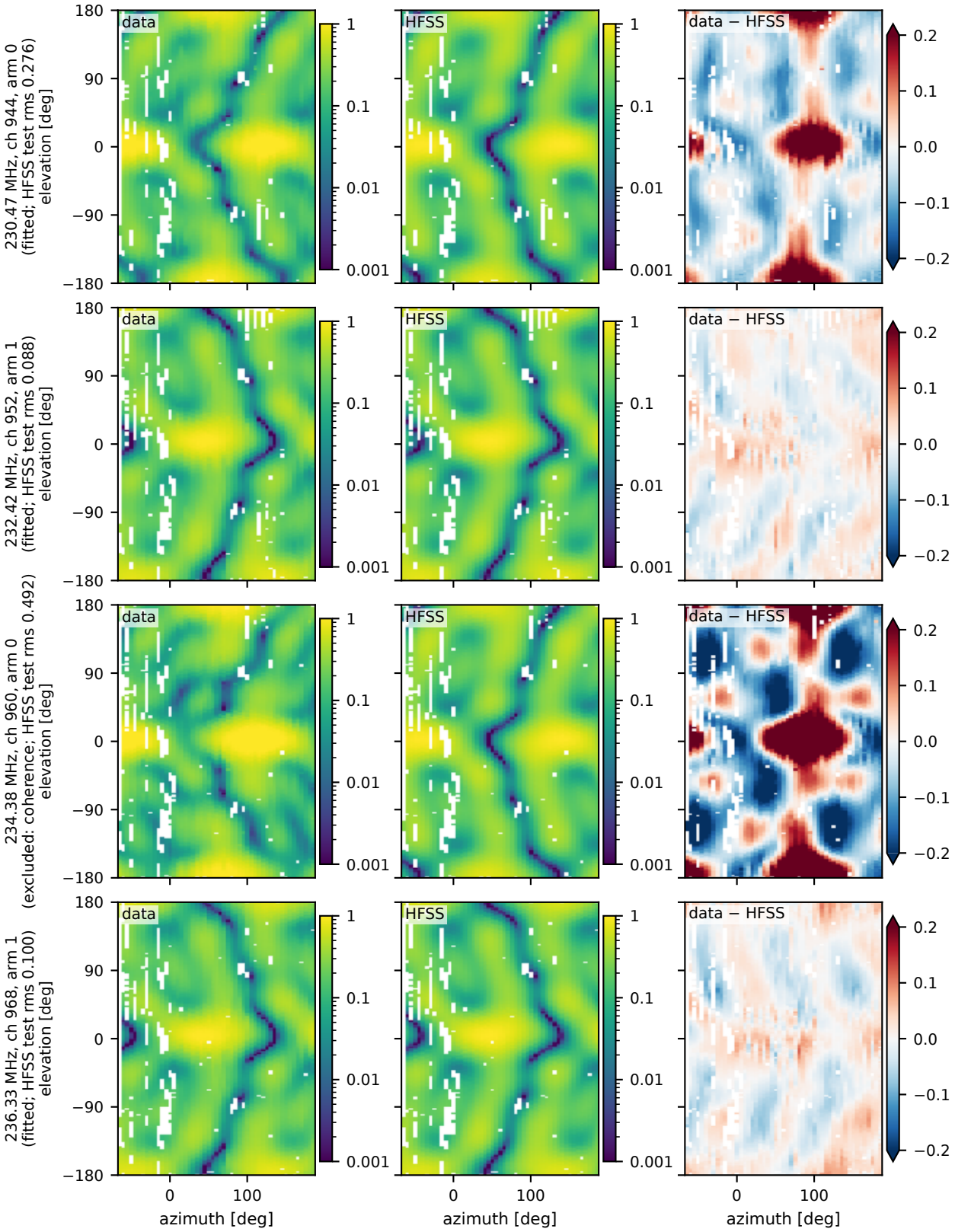


Figure 35: Raster maps, 230.47–236.33 MHz (page 24 of 26); panels as in Fig. 7.

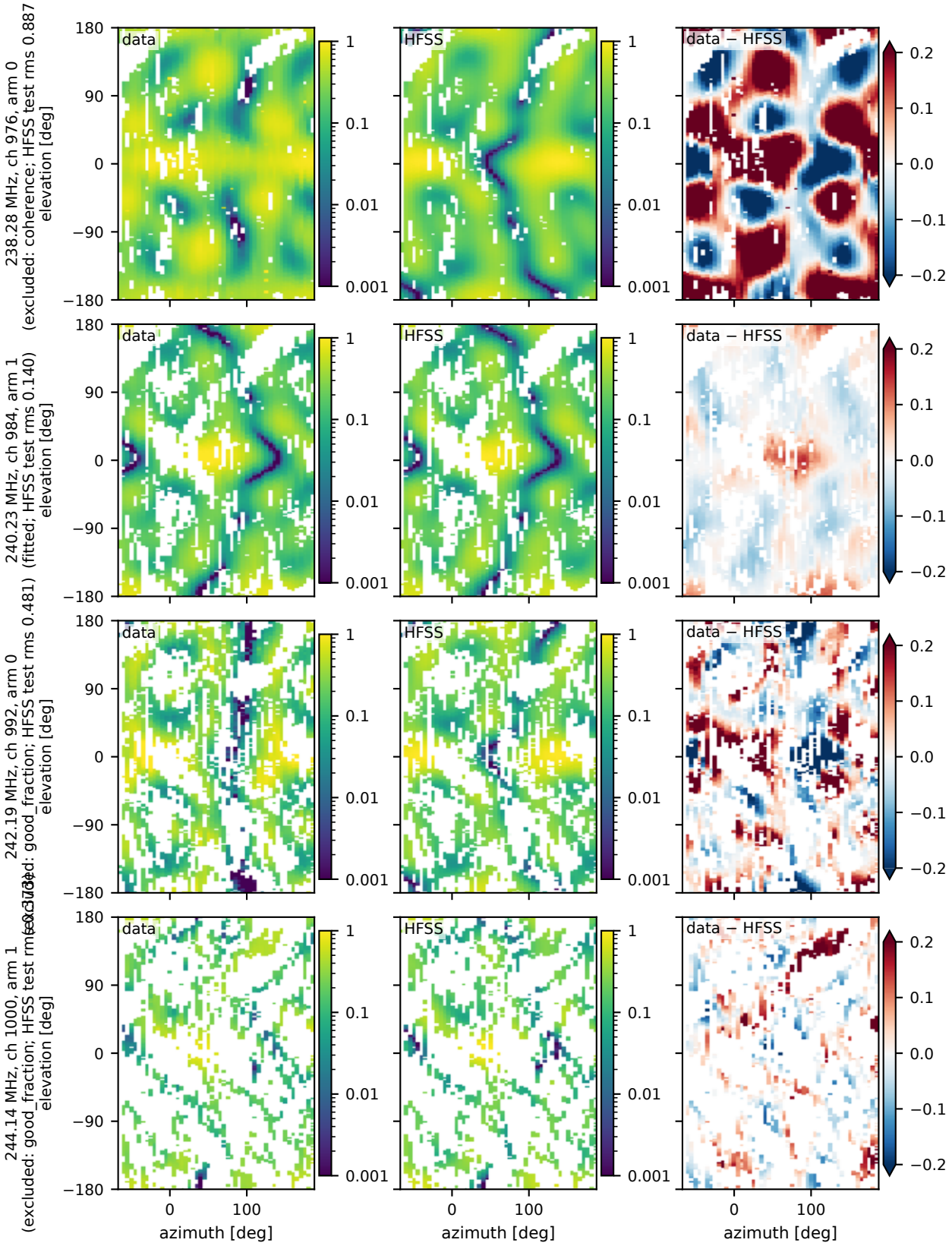


Figure 36: Raster maps, 238.28–244.14 MHz (page 25 of 26); panels as in Fig. 7.

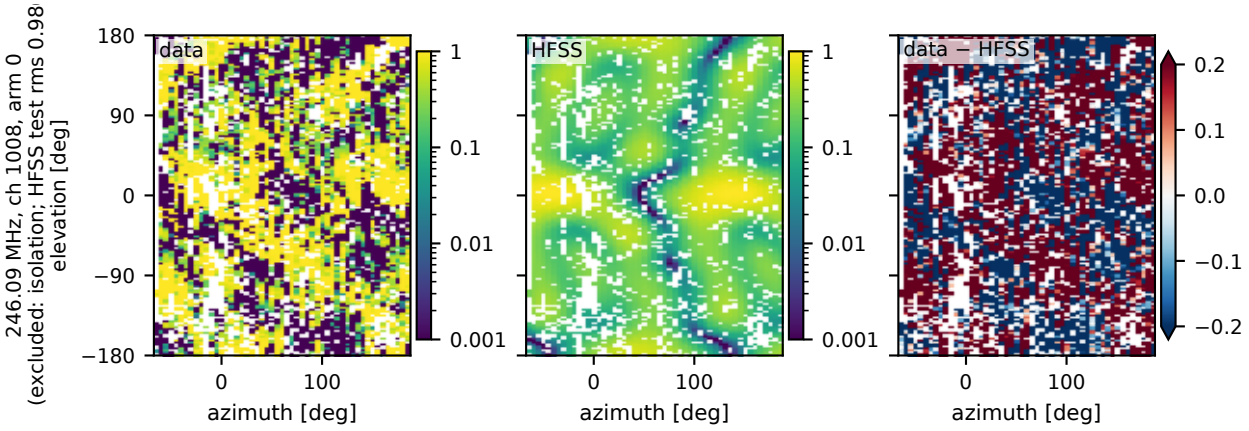


Figure 37: Raster maps, 246.09 MHz (page 26 of 26); panels as in Fig. 7.